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Modelling MEG-based resting-state functional connectivity to predict inter-individual differences in motor tasks

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Acronyms

AP Action Potential

AN Auditory Network

BLP Band Limited Power

CON Control Network

DAN Dorsal Attention Network

DMN Default Mode Network

ECG Electrocardiography

EEG Electroencephalography

EMG Electromyography

EOG Electrooculography

ERC European Research Council

FC Functional Connectivity

fMRI Functional Magnetic Resonance Imaging

FPN Fronto-Parietal Network

GCS Geometric Correction Scheme

HCP Human Connectome Project

IC Indipendent Component

ICA Independent Component Analysis

LN Language Network

MEG Magnetoencephalography

MN Motor Network

MNE Minimum-Norm Estimator

MUAP Motor Unit Action Potential

PSP Postsynaptic potential

RMS Root Mean Square

ROI Region of Interest

SERF Spin-Exchange Relaxation-Free

SQUID Superconducting Quantum Interference Device

VAN Ventral Attention Network

VFN Visual Foveal Network

VPN Visual Peripheral Network

ZCR Zero-Crossing Rate

Abstract

Individual characterization of subjects based on their functional connectome has become a highly coveted goal in contemporary neuroscience research. Recent studies have demonstrated an accurate identification of subjects across rest and task conditions, indicating that individual differences in brain connectivity act as a "fingerprint" to identify single subjects from a large group. This study is a contribution to the ERC *HANDmade* project (SH4, ERC-2017-STG) directed by Prof. Betti (University "La Sapienza" of Rome), which investigates how natural hand usage shapes behaviour as well as intrinsic and task-evoked brain activity. In particular, the thesis work aims at modelling magnetoencephalography-based resting-state functional connectivity to predict inter-individual differences in motor tasks. It also investigates the possibility of finding a relationship between functional connectivity data at rest and muscular activation. Results indicate that individual differences in task-evoked functional connectivity can be related to individual differences in spontaneous cortical activity (at rest), but the same cannot be assessed with muscle activation. This could be due to the simplicity of the model or the features used to represent the data. Therefore, further tests should be done in the future to overcome these limitations and open novel opportunities for future developments of robotic-assisted technology and neuroprostheses, based on personalized educational training programs.

Introduction

The human brain is characterized by inter-individual variability in neuroanatomy and physiology that is reflected in large individual differences in human cognition and behaviour. A fundamental question in brain research is what makes individuals different from each other. This question can be addressed at different levels of organization, starting from genetics or neurotransmitters, going through structural or functional measures of brain regions and networks, and ending at behavioural phenotypes or clinical outcomes [1].

In recent years, the increasing availability of public neuroimaging data has given rise to an increasing number of studies aiming to identify functional and structural differences that can explain pathologies and behaviour. A common approach to analyse brain organization in individuals is through the study of their functional connectomes, i.e. the brain network [1]. The functional connectome can be understood as the adjacency matrix of a brain network where nodes are neural regions of interest (ROIs) and edges represent functional interactions between those ROIs. Recent works on functional magnetic resonance imaging (fMRI) show that functional connectomes can serve as "fingerprints" of individual subjects [2] [3]. Specifically, Finn et al. [2] demonstrated that such connectivity profiles could be used to distinguish individuals across rest/task modalities. Tavor et al. [4] developed a machine learning algorithm to predict individual variability in task-evoked brain activity even for individuals with unique or unusual brain activation patterns. However, it is still unclear whether functional connectomes assessed at faster temporal scales, with magnetoencephalography (MEG), have specific properties comparable to those observed at slower temporal scales (assessed through

fMRI). Moreover, it could be interesting to investigate the possibility of finding a relationship between fingerprints extracted from brain connectivity and muscle activation during motor tasks.

This study aims to answer the abovementioned research questions. It is part of the ERC HANDmade project (StG, SH4, ERC-2017-STG; start date: 2018-02-01, end date: 2023-07-31) [5] whose principal investigator is Prof. Viviana Betti, Associate Professor with the University "La Sapienza" of Rome. The main purpose of the project is to test the hypothesis that the intrinsic brain activity represents and maintains an internal model of the environment built through the integration of information from visual and bodily inputs. In this framework, the hand has a special role, as it represents the primary means of interaction with the environment. In this thesis work MEG data acquired during resting-state and two different motor tasks, i.e. finger tapping and toes squeezing, were used to explore the extent to which individual differences in task-evoked functional connectivity (FC) can be predicted by differences in the brain connectivity at rest. FC matrices were estimated for a cohort of 51 subjects starting from MEG data from the Human Connectome Project (HCP). In the beginning, a consistency analysis was carried out to identify analogies and differences in functional connectivity data within and between subjects. Afterwards, a machine learning algorithm was developed to predict task-evoked FC and identify differences among subjects. Results are presented and analyzed for different brain networks and frequency bands in order to identify the set of conditions that provides the best estimate. Finally, a preliminary analysis investigates the possibility of finding a relationship between resting-state FC data and muscle activation during motor tasks, exploiting electromyographic (EMG) data.

This study shed some light on the idea that, in each individual, the dynamics of the spontaneous activity predicts the dynamics of brain activation during simple motor tasks. It can be used to investigate inter-individual differences between healthy and pathological populations, such as paralyzed patients or subjects affected by motor impairments [6]. Moreover, the discovery of MEG connectivity fingerprints may be used for the development of personalized educational training programs and clinical practices, improving rehabilita-

tion outcomes [2].

The thesis work is organized as follows:

- **Chapter 1 - Background.** The first chapter describes some basic concepts, necessary to understand the following analyses. It presents a general overview of the brain organization and its subdivision into functional networks. Afterwards, it provides some information about MEG and EMG acquisition process, together with a theoretical description of the regression techniques exploited in this work.
- **Chapter 2 - Methods.** It begins with a quick description of the HCP and the paradigm used to acquire the data. Then, it presents the pre-processing applied on MEG and EMG data and it describes the algorithms and the metrics used in this study. In particular, the architecture of the models and the metrics used for the evaluation of the performance are detailed in § 2.5 and § 2.6. From Section 2.7 onwards the discussion focuses on the analysis of EMG data, with a quick description of the different trials performed with them.
- **Chapter 3 - Results.** This chapter collects the main results obtained with the methods described in the previous chapter. In particular, it reports the results of the consistency analyses and the performance of the model applied to FC data.
- **Chapter 4 - Discussion.** In this chapter, are reported some observations derived from the results previously described.
- **Conclusions.** Finally, a summary of this thesis work is described, together with the model limitations and the future improvements.

Chapter 1

Background

1.1 Brain

The brain is one of the largest and most complex organs in the human body. It is contained in the cranium and is divided into three parts: the forebrain, which contains also the cerebrum, the midbrain and the hindbrain. The cerebrum is the largest part of the brain. It consists of two hemispheres, right and left, connected by bundles of nerve fibers that guarantee a continuous communication between the two parts. Both hemispheres are made up of an outer coating of grey matter, the cerebral cortex, consisting of nerve cells that receive information from the peripheral nervous system and guide the conscious movement of the body [7]. The surface of the cerebral cortex is extensively folded, forming ridges called *gyri* and valleys called *sulci*. The folding increases the surface brain areas and the number of neurons within the cerebrum, allowing for greater processing of information and cognitive abilities. The grooves between the folds divide each hemisphere into four lobes with different functions: the frontal lobe is located in the forward part of the brain, it is associated with reasoning, motor skills, higher-level cognition and expressive language; the parietal lobe, immediately behind the frontal one, is involved in the processing of body senses; the temporal lobe, in the lateral part of the head, is associated with hearing, memory and emotion; the occipital lobe, at the very back of the brain, is involved in interpreting

visual stimuli [8].

The main building blocks of the brain are neurons and glial cells. The latter provide neurons with nourishment, protection and structural support, but the fundamental cells for the processing of information are neurons. Neurons are excitable cells that can receive, process and transmit information to adjacent cells. A neuron consists of a soma which contains the cell nucleus, the dendrites and an axon (see Fig. 1.1) [9]. The dendrites receive signals

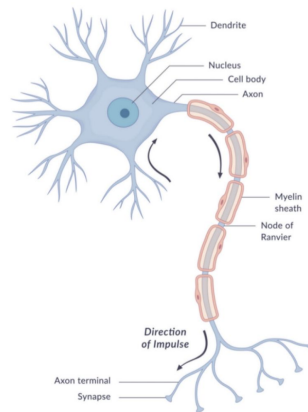


Figure 1.1: Neuron anatomy (modified from [10]).

from other nerve cells and integrate the inputs before transmitting them to the soma, while the axon carries out the output signal. The signals are transmitted in the form of electrochemical pulses called action potentials (APs). These pulses propagate across specialized junctions, the synapses, which acts as gateways for the transition of positive or negative ions across the cell membrane. As a result, the membrane potential of the neurons involved in the communication increases, and the AP is transmitted to the adjacent cell (Fig. 1.2). A single neuron may contain thousands of synapses, connecting it to many other cells. On average, the human brain contains about 100 billion neurons and more than 10^{15} synapses.

1.1.1 The brain network

Due to the high complexity of the brain structure, it has become useful to model and analyze the brain in terms of a network. The graph model of the

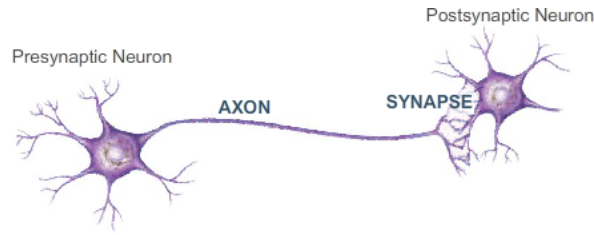


Figure 1.2: Communication between pre-synaptic and post-synaptic neurons (modified from [11]).

brain is an abstract mathematical model used to represent the relations between ensembles of neuronal elements, referred to as nodes [12]. The pairwise relations can represent the physiological activity between neurons (i.e. functional connections), or the anatomical connections between neuronal cells (i.e. structural links) [13]. Brain networks can be described at different spatial levels, such as microscale, mesoscale, and macroscale or large-scale. At the large-scale level, brain regions can be considered as nodes and functional connections between these regions as edges [14]. Going more into detail, the large-scale network can be subdivided into subsets of nodes highly connected between themselves with respect to the rest of the network. The spatial arrangement of these subnetworks includes sets of brain regions that are anatomically proximal to one another as well as sets of brain regions that are distributed across the brain. The reason these areas are highly connected is that they serve a common set of functions but in some cases, it is also possible that different networks work together to make more complex functions such as mathematics, speech etc. [15]. These networks can be identified at different resolutions and with different methodologies, therefore there is not a common atlas to define them [16]. Most of the time, the following 10 networks are considered (see also Fig. 1.3):

- auditory network (AN) plays an important role in the interaction with the environment [17];
- control network (CON) initiates and modulates cognitive control [16];
- dorsal attention network (DAN) is engaged in focused attention for

externally-directed tasks [18];

- default mode network (DMN) is involved in episodic memory processing for internal thinking and imagination [19];
- fronto-parietal network (FPN) is involved in sustained attention, complex problem-solving and working memory [20];
- language network (LN) participates in speech and language communication;
- motor network (MN) is activated during motor tasks;
- ventral attention network (VAN) is involved in reorienting attention when a salient stimulus is detected [21];
- visual foveal network (VFN) allows perception of fine-grained details in images [22];
- visual peripheral network (VPN) is involved in perceiving the broad spatial scene and selecting the attended regions for the fovea [22].

1.1.2 Functional connectivity analysis

In recent years, the importance of measuring connectivity between spatially separate but functionally related brain areas has become of key interest in the study of brain function [23]. Functional connectivity (FC) quantifies the statistical dependencies between the activity of brain regions. This activity can be measured with different neuroimaging techniques, such as electroencephalography (EEG), magnetoencephalography (MEG) or functional magnetic resonance imaging (fMRI). Frequently, FC is defined by measuring similarity between brain signals with Pearson’s correlation: if two signals from anatomically separated brain regions are correlated, the regions are functionally connected in the brain [23]. The level at which FC is measured can vary depending on the aim of the study. It can be measured between the

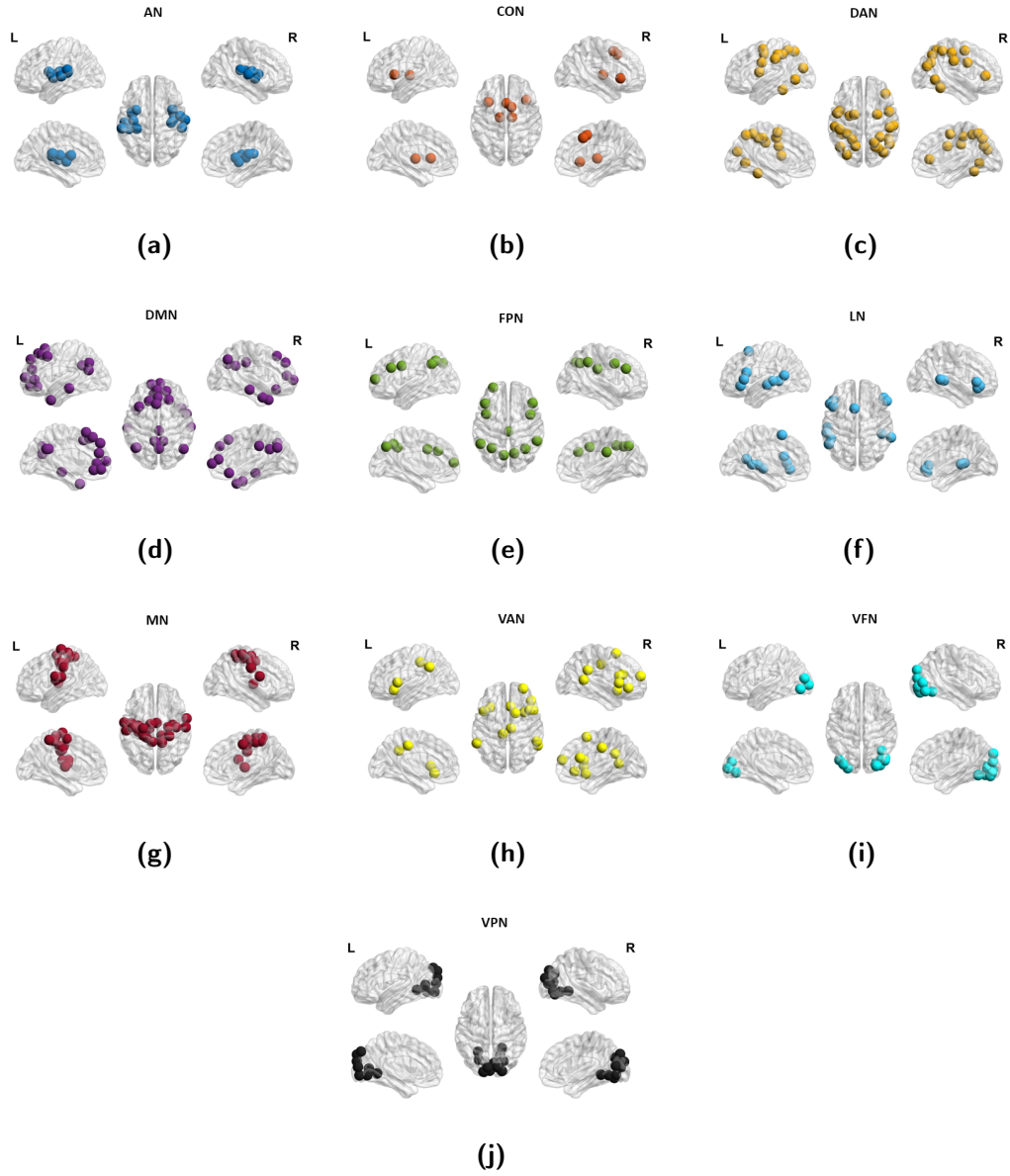


Figure 1.3: Brain networks: lateral, medial and dorsal view. (a) auditory network (AN), (b) control network (CON), (c) dorsal attention network (DAN), (d) default mode network (DMN), (e) fronto-parietal network (FPN), (f) language network (LN), (g) motor network (MN), (h) ventral attention network (VAN), (i) visual foveal network (VFN), (j) visual peripheral network (VPN). (Images created with *BrainNet Viewer* toolbox).

left and right hemispheric motor cortex; between regions belonging to different functional networks, or between single nodes, extracted with a specific parcellation scheme from the brain. The network representation of functional connectivity is referred to as a *functional connectome* and can be mathematically described by an adjacency matrix: each entry of the matrix reflects the edge strength or the functional connection between nodes in the corresponding row and column. This allows examining the brain network using graph analytical methods. In recent years, the use of graph theory on neuroimaging data has spread among the research community and many new graph organizational measures are designed and tested to examine the functional connectome of the brain and reveal important properties about its organization.

1.2 Magnetoencephalography

Magnetoencephalography (MEG) is a functional neuroimaging tool that measures the magnetic fields generated from neuronal activity in the brain. It is one of the most used techniques to detect the activity of the brain thanks to its non-invasiveness and its high temporal and spatial resolution. The first MEG recording was carried out by David Cohen in 1968 using a copper induction coil. The measurements were made in a magnetically shielded room. The coil detector was not sensitive enough, resulting in a poor and noisy MEG signal [24]. A few years later, Cohen used a superconducting quantum interference device (SQUID) detector, just developed by James E. Zimmerman [25], and a better shielded room. This time the measurements were clear, attracting the interest of neuroscientists from all over the world. In the 1990s, the introduction of high-density (200+) detector grids, covering the whole head, allowed to measure the activity of the brain rapidly and efficiently [26]. In recent years, many studies have been carried out to replace SQUID detectors with spin-exchange relaxation-free (SERF) magnetometers to improve the portability of the scanners. However, SQUIDs still remains the most used magnetometers for MEG devices.

MEG represents the activity of neuronal populations in which communication is accomplished via synchronized oscillatory activity during cognitive activity. The processing of information in the brain involves electric currents produced by the movement of sodium and potassium ions within and between neurons [27]. These currents sum up coherently with their orientation to produce a magnetic field that can be measured outside the head during MEG measurements. The source of the electrical currents of the brain, and hence the magnetic fields, lies in the neurons of the cortical grey matter. The transfer of information between neurons occurs at the level of synapses and involves the presynaptic neuron and the postsynaptic one. The presynaptic neuron stores neurotransmitters in vesicles at the axon terminal. When the presynaptic potential reaches the terminal of the axon, it triggers a release of neurotransmitters into the synaptic cleft. These neurotransmitters bind to the receptors in the dendrites of the postsynaptic neuron. As a result, the receptors change their shape and open up ion channels through the cell membrane, allowing specific ions to enter the cell [28]. The charged ions alter the transmembrane potential of the postsynaptic cell generating a postsynaptic potential (PSP). If the potential alters the equilibrium potential (-70 mV) positively or negatively, an excitatory or inhibitory PSP is generated. The local PSPs in the dendrites sum up at the root of the axon. If the overall potential reaches the firing threshold (-55 mV), an AP is generated and the neuron fires [29]. However, it is only the response of the PSP that can be detected outside of the brain during MEG recording. This is caused by several reasons. The presynaptic action potential is very fast (~ 2 ms) and it is biphasic, meaning that it has both a positive and a negative deflection, as shown in Fig. 1.4a. Therefore, spikes are not perfectly synchronized and their summation cancels each other out. The PSPs, instead, are slower (~ 10 ms) and monophasic allowing these deflections to add up more easily (Fig. 1.4b).

The synchronicity and the superposition also rely on the geometric configuration of the pyramidal cells. The MEG signal, indeed, doesn't measure all neural activity: only neurons tangentially oriented along the skull surface generate magnetic fields that can be measured on the cortex. When a sufficient number of neurons (in the order of 10^5) fire at about the same time

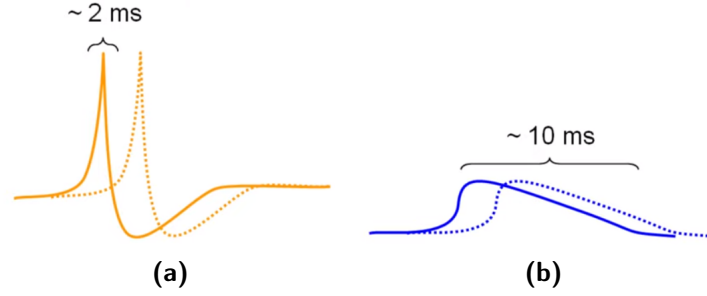


Figure 1.4: Presynaptic potential (a) and excitatory postsynaptic potential (b) (modified from [30]).

and with a similar orientation, the electrical currents sum up, generating a magnetic field detectable by MEG. The generated neuromagnetic field is extremely small and it diminishes as $1/r^3$ with the distance of r [27]. As a consequence, only magnetic fields generated along the cortex can be detected, thanks to superconducting magnetometers.

The typical amplitude of a MEG signal is in the order of femtotesla ($1 \text{ fT} = 10^{-15} \text{ T}$) with a frequency band of $[0.5 - 1000] \text{ Hz}$, even if $[1 - 80] \text{ Hz}$ is the most typical range. The temporal resolution is around 1 ms, allowing measurement of very fast time-frequency dynamics of brain networks. Also the spatial resolution is high with respect to other techniques used for measuring brain activity. Indeed, MEG fields pass through the head without any distortion providing a source localization with millimeter precision in the cerebral cortex [31].

The frequency band of the MEG signal can be divided into different brain rhythms: delta (δ), theta (θ), alpha (α), beta (β), gamma (γ), as shown in Fig. 1.5. Delta waves lie within the range of 0.5-4 Hz and they are primarily associated with deep sleep [32]. Theta waves, in the range of 4-8 Hz occurs during unconscious states or during meditation and creative inspiration [33]. Alpha waves mostly appear over the occipital region of the brain. They are between 8 and 15 Hz and they indicate a relaxed awareness without any attention or concentration [34]. Beta waves mostly appear in the central and frontal region of the brain and they are usually between 15-35 Hz. However, the frequency range can be further divided into beta low (15-25 Hz) and

beta high rhythms (26-35 Hz). The former is thought to be associated with states of concentration and problem solving, while the latter is more engaged in highly complex thought and integration of new experiences [35]. Gamma waves, with frequencies above 35 Hz, play a role in a relatively late stage of motor control, encoding information related to limb movement rather than to muscle contraction [36].

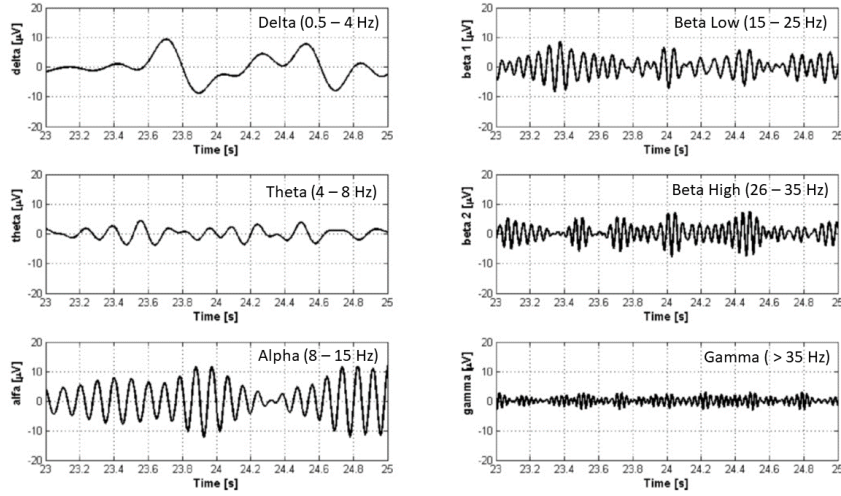


Figure 1.5: Brain rhythms (modified from [37]).

1.2.1 Instrumentation and data acquisition

As stated in the previous section, the firing of a single neuron is insufficient to produce a measurable magnetic field. However, even a large population of neurons can still create only weak magnetic currents, thus requiring highly sensitive sensors as well as significant noise reduction from the environment and other sources of disturbances to measure it. The sensor used to measure these minute magnetic fields is the SQUID, a highly sensitive magnetic field meter with a sensitivity of $3fT/\sqrt{Hz}$. SQUIDS are based on the phenomena of superconductivity that is the state in which electrical resistance is zero. This phenomenon to happen requires ultra-cold temperatures. For this reason, SQUID sensors are immersed in a dewar filled with liquid Helium and operate at temperatures of $4K$ or $-269^{\circ}C$. A SQUID sensor converts the

magnetic flux into voltage such that alterations in the magnetic flux result in voltage changes across the SQUID circuitry. Due to its small size, it is used together with a flux transformer to increase the sensitivity. The flux transformer can be constructed to measure the magnetic field itself (i.e. magnetometer) or the gradient of the magnetic field (i.e. gradiometer) to cancel out unwanted sources of interference like the magnetic fields generated by muscle artifacts, by the heart or by other internal organs [26]. For MEG purposes, magnetometers or gradiometers are uniformly distributed in the inner surface of a helmet-shaped dewar which covers the entire head, as in Fig. 1.6. Their number can vary between 100 to 300 allowing the simultane-



Figure 1.6: MEG dewar (modified from [38]).

ous recording of signals all over the scalp. The subject's head is positioned under the dewar as close as possible to the sensors, but most of the time it is not spatially constrained. The consequence is that the subject's head, and thereby the neuronal sources in the brain generating the neuromagnetic fields, can move relative to the MEG sensors [39]. To minimize the effects of head movements, subjects are instructed to maintain the same posture during the recording. Furthermore, some devices allow continuous head location monitoring for online and offline head movement compensation [40].

There are also other interferences that can affect the recorded signal: system-related artifacts, due to SQUID jumps, noisy, broken or saturated sensors; external artifacts, arising from generators of magnetic fields outside the human body such as magnetic noise from power lines; physiological artifacts, caused by eye movements, eye blinks, cardiac and muscular activity [41]. The physiological artifacts can be partially suppressed by recording some external signals (e.g. EOG, ECG and EMG) during the MEG measurement, in order to facilitate the identification of artifactual components. Another source of noise comes from the external environment. To reduce the background noise, the measurements are performed in a magnetically shielded room. Its walls are made of a combination of multiple layers of mu-metal (an alloy consisting mostly of nickel and iron) and a single layer of aluminium. The first material provides an effective ferromagnetic shielding against low frequencies of urban magnetic noise, while the latter protects against high frequencies [27].

1.3 Electromyography

EMG is a technique for evaluating and recording the electrical activity produced by skeletal muscles. It is of particular relevance in sports science and rehabilitation medicine but also to identify neuromuscular diseases. Historically, EMG can be dated back to 1791 when Luigi Galvani recognized the relationship between the electrical stimulation of a nerve and the contraction that is generated in a muscle. The first recording of this activity was made in 1890 when the voluntary contraction of a muscle was detected by Emil Du Bois Reymond [42]. However, this signal was used for clinical purposes only after World War II [43].

The EMG signal provides insights into the neural activation and dynamics of the muscles. The functional unit of muscle contraction is the *motor unit*. It consists of a motor neuron, connected to the spinal cord, and to all skeletal muscle fibers it innervates (see Fig. 1.7) [44]. When a motor unit fires, the impulse, called *action potential*, is carried by the motor neuron to the muscle through the neuromuscular junction, or the motor endplate. As a result, an

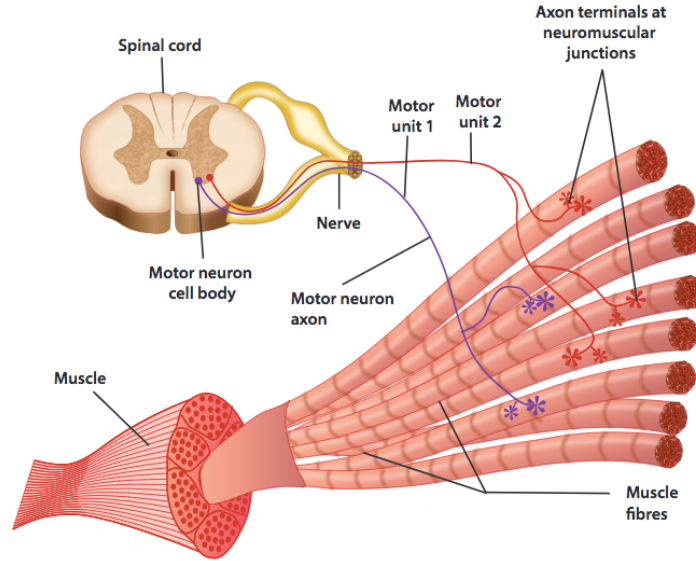


Figure 1.7: Motor unit (modified from [45]).

action potential is elicited in all innervated muscle fibers of that motor unit and the muscle contracts. The sum of all the electrical activity required for muscle fibers contractions is known as a motor unit action potential (MUAP) and it is measured during an EMG recording. The EMG signal measures the net contribution of all active motor units detectable under the electrode site. The number of active motor units is related to the contraction's strength of the muscle. In particular, the level of muscular contraction depends on two mechanisms: the time recruitment which is strictly related to the firing frequency of the motor units, and the spatial recruitment that depends on the number of motor units active in the portion of the muscle under consideration [46]. These two mechanisms influence the magnitude of the observed signal.

1.3.1 Signal acquisition

The acquisition of EMG signals is carried out by electrodes that acquire the electrical potential difference between two locations above the contracting muscle. They can be positioned directly in contact with muscle fibers, i.e. needle electrodes, or on the skin, i.e. surface electrodes [47]. Needle

electrodes can acquire EMG signals from a single muscle fiber with high precision, but their invasive nature is a major drawback, making them unviable for any commercial application. Surface electrodes are positioned on the skin, at the belly of the muscles, and they acquire EMG signals from several motor units, obtaining noisy signals [46] [48]. Application of surface EMG electrodes requires proper skin preparation beforehand. To obtain an EMG signal of good quality, the skin impedance must be considerably reduced. For this purpose, the dead cells on the skin must be completely removed from the location where the EMG electrodes are to be placed. Furthermore, the skin should be cleaned with alcohol to eliminate any wetness or sweat on the skin [47].

To increase the SNR, the EMG electrodes should be placed along the longitudinal midline of the muscle. The frequency content of the signal ranges from 6 to 500 Hz, even if the typical frequency power is between 20 and 150 Hz. The amplitude is very low ($\pm 5mV$) and when the muscle is relaxed a noise-free EMG baseline with an amplitude around $3 - 5\mu V$ can be observed. The investigation of the EMG baseline quality is very important to understand the level of noise in the recording and to avoid misleading interpretation during muscle contraction [49]. Indeed, EMG signals are highly affected by noise. The most common type of noise is due to the instability of the electrode skin interface and the movement of the electrode cable, but it can be partially suppressed thanks to a differential amplifier characterized by high input impedance and very low output impedance [50]. Despite this pre-amplification stage, the signal is still weak and highly affected by various types of noise like power line interference, movement artifacts, cross-talk interference due to monitoring of undesired muscular activity, and ECG or other biological activity artifacts [51]. Therefore, EMG signals need to be properly filtered and amplified for further analysis.

1.4 Regression models

Machine learning is one of the most important branches of artificial intelligence, specifically for engineering and data scientists [52]. It is a very wide field that can be applied to solve many different problems. One of the most common models of machine learning is regression. Regression analysis is a modelling technique that investigates the relationship between a dependent (target) and independent variable (predictor). This technique is used for forecasting, time series modelling and finding the cause-effect relationship between variables [53]. The standard approach consists of selecting the model type, based on prior knowledge about the data, and then choose the suitable method to estimate the model parameters. Given X and Y the independent and dependent variable respectively, b the unknown parameters and ϵ the error term, the regression model can be written as follows:

$$Y = f(X, b) + \epsilon \quad (1.1)$$

The goal is to estimate the function $f(X, b)$ that most closely fits the data, providing an error term as small as possible [54]. The form of this function is based on knowledge about the relationship between Y and X . Once the model has been determined, different tools can be used to estimate the parameter b . For example, least-squares approach aims at finding the value of b that minimizes the sum of squared error between the dependent variable and the prediction, i.e. the model output, $\hat{Y} = f(X, \hat{b})$ [55].

Depending on the problem specification, different kinds of regression analysis can be applied. Simple linear regression involves the following model

$$f(X, b) = b_0 + b_1 X_1 + \epsilon \quad (1.2)$$

Due to the simplicity of the model, the predictions obtained with simple regression can be rather inaccurate. Multiple linear regression models are preferred when the response variable can be predicted through a linear combination of explanatory variables and parameters [56]. The mathematical

representation of multiple linear regression is

$$Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + .. + b_tX_t + \epsilon \quad (1.3)$$

A less common type of regression analysis is nonlinear regression that is used with complicated datasets when the model function is not linear in the parameters [56].

Most of the time researchers are also interested in assessing the predictive power of the model. When the size of the dataset used for training the model is limited, one common technique is to use cross-validation. Essentially, cross-validation splits the dataset into K subsets. Iteratively, each fold is used as a test set, while the others together constitute the training set. The model is fitted and evaluated K times, and the prediction error from each model fitting is averaged to determine the expected prediction error for the whole model [57]. The choice of the number of splits (or "folds") of the data is not trivial: a high number of folds could lead to computational problems, while a low number could lead to a high bias. Usually, $K = 3, 5$ or 10 is selected [58]. Additionally, leave-one-out cross-validation (LOOCV) is a particular case in which K is set equal to the number of examples in the dataset. Due to the fact that each fold consists of only one sample, the computational cost of this type of validation is very high, but it provides a robust estimate of model performance [58] [59]. Another way to implement LOOCV is to keep the data from one subject out, while applying the model with the remaining subjects, and use the fitted model to make prediction for the out-of-sample subject [4].

In this thesis work, this latter approach was applied, together with regression models, to identify a relationship between resting-state and task-evoked FC data and to identify single subjects in the dataset.

1.5 Graph theory

Graph theory is a branch of mathematics that dates back to the 18th century. Today, applications of graph theory can be found in all scientific disciplines

because they can offer insights into the structure and function of networked systems, including their architecture, evolution and development [60].

A graph is a mathematical representation of a network. It is usually indicated with $G = (V, E)$, where V is the set of vertices (also referred to as nodes) and E is the set of edges. A graph without loops and with at most one edge between any two vertices is called a *simple graph*. Simple graphs can be classified as directed or undirected, weighted or unweighted. A graph is directed and weighted if the edges have a specific direction (from the source to the end node) and are associated with a number indicating the strength of the connections. The topology of a graph is represented by a $N \times N$ adjacency matrix $\mathbf{A}$ where N is the number of nodes and a_{ij} are the entries of the matrix that reflect the presence or absence of a connection between the vertices [12]. By exploiting the matrix representation of the network, several graph theoretical measures can be computed to capture important aspects of global and local connections. Global measures are primarily aimed at revealing segregation and integration of information flows within networks and resilience against failure. Local properties, instead, capture information about specific nodes in the network [61]. For example, an important measure associated with each node is its *degree* which is defined as the number of edges connected to that vertex, in the case of an unweighted network, or the sum of the edges' weights, in a weighted graph. Another important concept in graph theory is the *path*: a sequence of adjacent edges, without repetition, that identifies a walk from one node to another. Despite the large number of measures that can be applied on a graph, it is important to carefully select the most suitable ones to capture the salient properties of the networks.

In this thesis work, graph theory was applied to the human brain at the scope of investigating the organization of this complex network. Different graph metrics were extracted from the adjacency matrix of the brain graph and tested to capture a relationship between brain connectivity and muscle activity during motor tasks.

Chapter 2

Methods

2.1 Human Connectome Project

The Human Connectome Project (HCP) is a five-year project sponsored by the National Institutes of Health (NIH), split between two consortia of research institutions: the *WU-Minn consortium* and the *MGH-UCLA consortium* [62]. The primary goal of the HCP is to acquire and share data about the structural and functional connectivity of the human brain and to provide these data as public resources for biomedical researches. In particular, the WU-Minn HCP consortium aims to characterize human brain connectivity and function in a population of 1200 healthy adults and to enable comparisons between brain circuits, behaviours, and genetics at the level of individual subjects. HCP subjects are drawn from a population of adult twins and their non-twin siblings, in the age range of 22-35 years [63]. The WU-Minn consortium collected a large amount of data from the participants, comprising fMRI and MEG data acquired during resting-state and different tasks like motor, working memory and language tasks. These data were shared with the scientific community in an open-source web-accessible neuroinformatics platform called *ConnectomeDB*. Importantly, these include unprocessed ("raw") imaging data and data after multiple levels of processing, needed for data analysis and interpretation.

In this thesis work, the focus is directed on MEG data, acquired during

resting-state and motor tasks. MEG data were collected on a whole head MAGNES 3600 scanner system (4D Neuroimaging, San Diego, CA) housed in a magnetically shielded room, located at the Saint Louis University medical campus. The MEG system includes 248 magnetometer channels together with 23 reference channels. Data were recorded at a sampling rate of 2034.5 Hz during three runs of resting-state and two runs for each task. In the motor tasks, participants were presented with visual cues suggesting the movement to perform: finger tapping, with left or right hand, and toes squeezing, with left or right foot. The paradigm for each run of the motor task is shown in Fig. 2.1. It consists of 32 movement blocks (8 for left hand, 8 for right hand, 8 for left foot and 8 for right foot). Each block started with an instruction screen, indicating the side (left, right) and the limb (hand, foot) to be used by the subject in the current block. Then, 10 pacing stimuli were presented in sequence, each one instructing the participant to make a brisk movement. Moreover, 10 interleaved blocks were used for fixation and indicated by a + sign in Fig. 2.1.

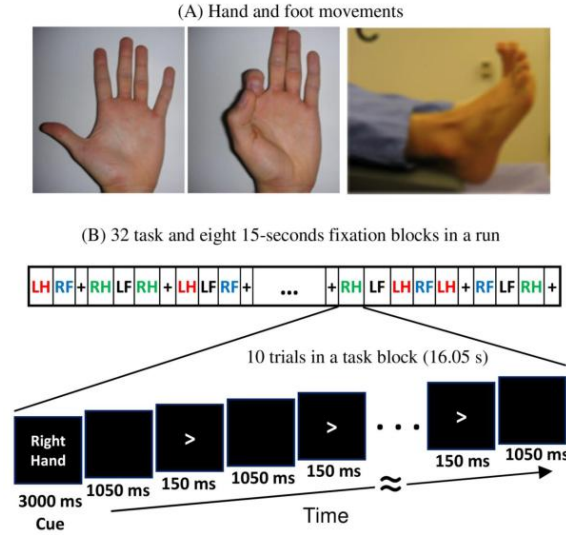


Figure 2.1: (A) Hand and foot movements during motor task, (B) Paradigm in a run of motor task (modified from [64]).

In addition to the recorded MEG channels, also the EMG activity was recorded for each limb. The main purpose of EMG signals is to measure

electrical activity produced by skeletal muscles during each scan of motor task and to define the onset of an event for hand and foot movements. EMG electrodes were applied to the participant’s skin on the lateral superior surface on the extensor digitorum brevis muscle (Fig. 2.2a) and on the first dorsal interosseus muscle (Fig. 2.2b) [64]. Data were acquired during the two runs of each motor task and then stored in the same format as MEG data, without any pre-processing operations.



Figure 2.2: EMG electrodes’ position during foot (a) and hand (b) movements (modified from [64]).

2.2 MEG pre-processing

MEG data for 51 subjects¹ from the HCP, were downloaded from the *ConnectomeDB* website and processed by the *Cognitive and Systems Neuroscience lab (Cosync lab)* [65] in order to obtain a FC matrix for each run.

The processing pipeline consisted in the removal of noise and artifacts by means of band-pass ($[1.3 - 150]$ Hz) and notch filters ($[59 - 61]$ Hz and $[119 - 121]$ Hz). Channels and signal segments contaminated by large artifacts were automatically identified by an independent component analysis (ICA)-based approach (using FastICA [66]) and removed for further analyses. Subsequently, the same ICA approach was applied again to separate environmental and physiological (e.g. cardiac, ocular) artifacts and brain independent components (brain ICs) from sensor space MEG signals [67]. The resulting

¹105923, 106521, 108323, 109123, 113922, 116726, 133019, 153732, 156334, 162026, 162935, 164636, 169040, 175237, 177746, 185442, 189349, 191841, 198653, 204521, 205119, 212318, 212823, 221319, 250427, 255639, 257845, 287248, 293748, 353740, 358144, 406836, 559053, 568963, 581450, 599671, 601127, 660951, 662551, 667056, 679770, 680957, 706040, 707749, 725751, 735148, 783462, 814649, 891667, 898176, 912447

ICs were projected into a source space of 8004 vertices by means of Tikhonov-regularized Minimum-Norm Estimator (MNE). A leakage-correction approach was applied to the IC source-space maps in order to remove the spurious co-dependence of the reconstructed sources, using the geometric correction scheme (GCS) approach [68]. Brain regions were parcellated into 164 ROIs, consisting of vertices belonging to the 10 networks described in Section 1.1.1. The band limited power (BPL) was estimated for each seed source and the vector activity filtered into three frequency bands: alpha [8-15 Hz], beta low [15-26 Hz] and beta high [26-35 Hz]. The filtered BLP timecourses were then used to build a FC matrix: a 164x164 symmetric matrix where each entry represents the FC strength between pairs of nodes. Values of FC were computed with Pearson's correlation (see Section 2.2.2) if the distance between two nodes was bigger than 35 mm [69], otherwise a "not a number" (NaN) value was assigned to avoid the co-dependence of closer nodes. In this study, the original 164x164 FC matrix was processed to extract only nodes belonging to the network of interest. Since the FC matrix is symmetric, only its upper triangular elements were considered and a link was retrieved for each couple on nodes. The number of nodes for each network is reported in Table 2.1 together with the number of connections.

Table 2.1: Brain networks tested in this thesis work, together with the corresponding number of nodes and links.

Network acronym	Network full name	#Nodes	#Links
ALLNTW	All Networks	164	13366
AN	Auditory Network	15	105
CON	Control Network	6	15
DAN	Dorsal Attention Network	24	276
DMN	Default Mode Network	28	378
FPN	Fronto-Parietal Network	12	66
LN	Language Network	13	78
MN	Motor Network	23	253
VAN	Ventral Attention Network	14	91
VFN	Visual Foveal Network	12	66
VPN	Visual Peripheral Network	17	136

The same analysis was repeated for each subject, for each run of rest and task conditions and each frequency band (α , β_{low} and β_{high}).

2.2.1 Degree-normalization and graph metrics

In several FC studies, the human brain is considered as a complex network, where N nodes represent neural elements or brain regions, and links correspond to the physical or functional connections among nodes. Links are usually undirected and weighted, reflecting the strength of correlation between pairs of vertices. In this framework, the FC matrix can be seen as the adjacency matrix of the brain graph. Since the brain network is highly interconnected, it is common to apply a normalization operation on its functional connectome to reduce the influence of highly connected nodes (i.e. hubs) with respect to others.

In this work, a *degree-normalization* operation was applied on the adjacency matrix of each subject [6]. The degree-normalization of $\mathbf{FC}$ was computed as follows:

$$\mathbf{FC}_{norm} = \mathbf{D}^{-1/2}(|\mathbf{FC}|)\mathbf{D}^{-1/2} \quad (2.1)$$

where $\mathbf{D}$ is the $N \times N$ matrix containing the degree values on its diagonal and zeros elsewhere. The resulting matrix $\mathbf{FC}_{norm}$ is symmetric and corresponds to the adjacency matrix of the normalized brain graph, where any excessive influence of nodes has been modulated by their corresponding weighted degree [70]. It is worth noting that degree-normalization on functional connectomes would potentially involve negative node degrees which would in turn generate complex entries in the normalized FC matrix. For this reason, degree-normalization was applied only to unsigned connectome, obtained considering its absolute value [6].

Moreover, different graph metrics were computed on the normalized matrix to extract meaningful topological properties of the graph and to investigate the presence of a relationship with muscular activity. The degree d_i of node i is defined as the sum of the weights of its neighbouring edges [71] that in the case of the functional connectome, correspond to FC values. To obtain

a global measure, the *average degree* (WD) is computed as follows:

$$WD = \frac{\sum_{i=1}^N d_i}{N} \quad (2.2)$$

where N is the number of nodes in the network [72]. Another important measure to capture the complexity and the efficiency of the network is the *average path length* (PL). It represents the shortest average distance between all pairs of nodes in the network and it is calculated with the following equation:

$$PL = \frac{2}{N(N-1)} \sum_{i,j} s_{i,j} \quad (2.3)$$

where $s_{i,j}$ is the shortest path length between i and j , defined as the minimum number of links traversed to get from node i to j [73] [74]. One way to compute the length of the shortest path in a weighted network is to use the Dijkstra algorithm: it sums the weights of connections from the source to the arrival node and it finds the path of least cost [75]. Finally, the *global efficiency* (E) is a measure of how efficiently is the exchange of information across the whole network. It is computed with the formula below

$$E = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{s_{ij}} \quad (2.4)$$

where N is the total number of nodes in the network and d_{ij} is the length of the shortest path between nodes i and j .

2.2.2 Pearson's correlation

Pearson's correlation is one of the most common types of correlation. The Pearson's correlation coefficient is an indicator of the strength of the linear relationship between two different variables and it is computed dividing the covariance of the two variables, x and y , by the product of their standard deviations, as follows:

$$r_{xy} = \frac{cov(x, y)}{\sigma_x \sigma_y} \quad (2.5)$$

Its value ranges between -1 and 1: a correlation of -1 indicates a perfect negative correlation, a correlation of 1 shows a perfect positive correlation, while 0 indicates no correlation at all. Typically, a hypothesis test is performed to verify the significance of the correlation estimated from the experimental sample for the entire population. To do so, the p-value is calculated and compared with a significance level α , usually equal to 0.05. The p-value reflects the probability that the correlation between the two variables occurred only by chance. If $p\text{-value} < \alpha$ the correlation is statistically significant, otherwise there is no significant linear correlation in the population [76] [77].

2.3 EMG pre-processing

EMG signals were contaminated by various noise or artifacts that originate at the skin-electrode interface, in the electronics that amplify the signals and in internal and external sources. Therefore, an initial pre-processing operation was needed to filter out the maximum amount of noise retaining as much of the desired EMG signal as possible. Fig. 2.3 shows unprocessed EMG signals in time domain and frequency domain considering the first run of motor task for an exemplary subject. First of all, a band-pass Butterworth

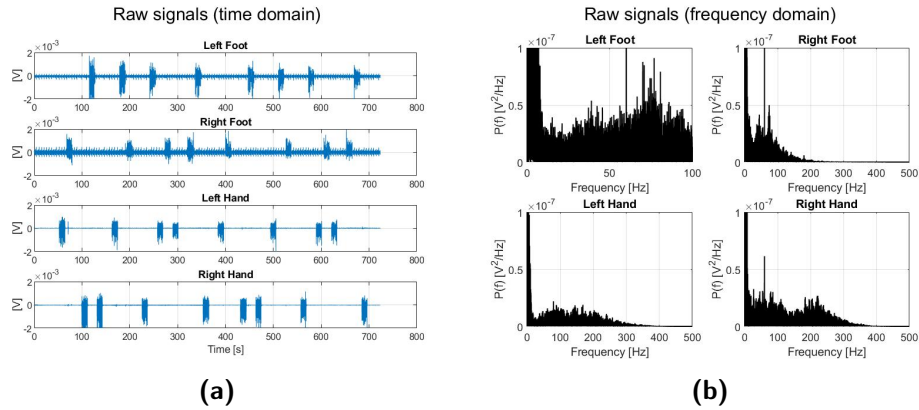


Figure 2.3: Raw EMG signals for left foot, right foot, left hand, right hand during the first run of motor task of the subject "106521". (a) Plots in time domain, (b) power spectral density estimation.

filter of order 4, with cut-off frequencies of 15 Hz and 450 Hz, was applied to retain only the frequency band related to EMG activity. The low frequency cutoff removes baseline drift, mainly associated with slow movements, electrocardiogram artifact and sweating, while the high frequency cutoff removes high frequency noise. However, as displayed in Fig. 2.4b, EMG signals still

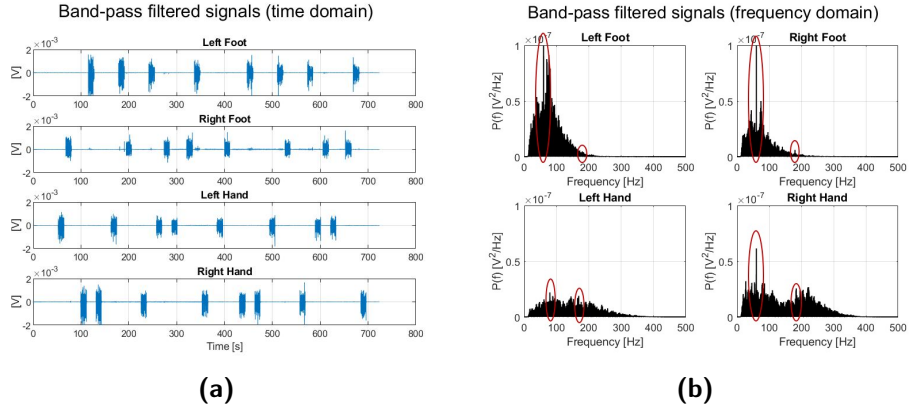


Figure 2.4: Time-domain (a) and frequency-domain (b) EMG signals after the band-pass filtering operation for left foot, right foot, left hand, right hand during the first run of motor task of the subject "106521".

appeared to be affected by large artifacts around 60 and 180 Hz. This was due to the powerline interference (PLI) generated by the linear alternative 60 Hz sinusoidal current from the power grid and its third harmonic. Therefore, two notch filters with cutoff frequencies of 60 and 180 Hz were applied to EMG signals and the filtered signals are shown in Fig. 2.5a in time domain and in Fig. 2.5b in frequency domain.

After these pre-processing steps, a further cleaning operation was performed to remove signal segments classified as "bad" during MEG processing. "Bad segments" consisted in short time epochs where internal or external noise sources had generated large magnetic fields, affecting the MEG signal [64]. To remove these epochs also from EMG signals, the latter were decomposed into segments lasting 800 ms, each one associated with one motor movement, and signal epochs previously rejected during MEG data pre-processing, were discarded from the EMG dataset, too. In this way, EMG

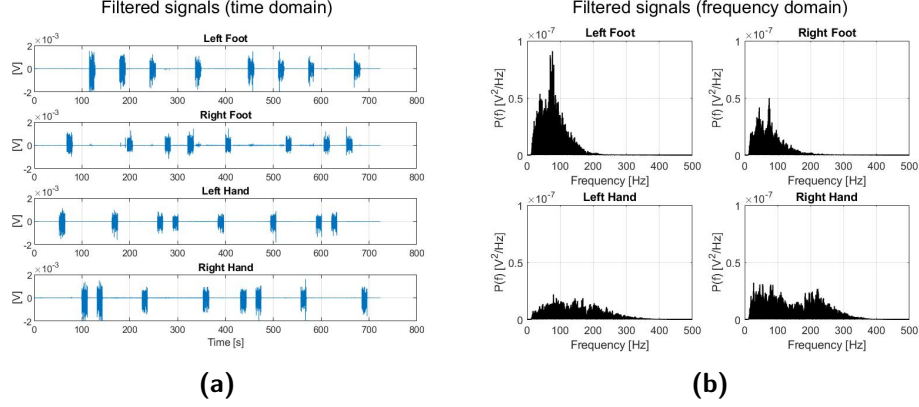


Figure 2.5: Filtered EMG signals in time domain (a) and frequency domain (b) during a single run of motor task for subject "106521".

signals were aligned with MEG ones in order to use them in the following analysis to investigate the interactions between brain and muscles.

2.3.1 Time-domain features

To capture the most significant properties of EMG signals, time domain features were extracted from EMG data [78]. Time domain features are in general easy to extract and analyze, as they are computed based upon the input signals amplitude. Some examples are the root mean square (RMS) and the zero-crossing rate (ZCR). RMS is a frequently used time-domain feature that represents the level of activity in the muscle during contraction. It is computed as follows:

$$RMS = \sqrt{\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} [f(t)]^2 dt} \quad (2.6)$$

where T_1 and T_2 are respectively the time limits of the signal segment considered in the computation, while $f(t)$ indicates the EMG signal in time domain. The ZCR is defined as the rate of sign changes of the signal during an interval of time. It reflects the characteristics of the muscle activities in the state of

muscle fatigue and it is computed with the following equation [79] [80]:

$$ZCR = \frac{\sum_{i=1}^{N-1} s(x_i, x_{i-1})}{N} \quad \text{with } s(z, y) = \begin{cases} 1 & \text{if } (z \cdot y) < 0 \\ 0 & \text{otherwise} \end{cases} \quad (2.7)$$

2.4 Similarity in functional connectivity

The functional connectome represents a mathematical way to characterize individual differences in connectivity patterns across brain regions. It has been demonstrated that FC matrices show characteristic patterns to each individual that can be detected across time [2] [4]. This uniqueness and stability allow identifying individuals via their functional connectomes [81]. Therefore, it is important to investigate the consistency of FC matrices both within and between subjects. In the former case, the consistency between FC matrices can be assessed using cosine similarity, while for the latter it can be interesting to evaluate the similarity in the matrix structure among different subjects.

2.4.1 Cosine Similarity

Cosine similarity is a similarity measure used to determine the level of similarity between two matrices considering the values of FC [82]. Suppose that X_i and X_j are two vectors formed by concatenating the upper triangular elements of the NxN functional connectivity matrices for subject i and j , respectively. The cosine similarity measure is given by the inner product of the two vectors divided by their norms, as reported in the formula below.

$$CS_{i,j} = \frac{X_i \cdot X_j}{||X_i|| ||X_j||} \quad (2.8)$$

The values of cosine similarity can vary from -1 (maximal dissimilarity) to 1 (perfect similarity).

In this study, this metric was used to assess the similarity value among resting-state FC matrices. For each subject, three CS values were computed

considering pairs of FC matrices during the three runs of resting-state. Then, an average operation was performed to obtain a single value of similarity for each subject, called CS_{tot} . Furthermore, in order to compare the results for different networks and frequency bands, the similarity values of all subjects were averaged to obtain the mean value of similarity, CS_{avg} , in a specific condition.

2.4.2 Structure Similarity

One major confound when estimating connectivity in source-reconstructed MEG is the *spatial leakage*. It leads to an estimation of spurious, blurred connections that may affect the connectomes' properties [83]. For this reason, "not a number" elements, i.e. NaN values, were used in this study for masking connectivity values when the nodes are too close to each other or when a leakage-correction was applied during MEG source reconstruction. The position of NaN values in the functional connectome was sometimes different between subjects. Therefore, between-subjects similarity was assessed by looking at the structure of the functional connectomes in order to reveal the level of similarity between subjects. At this scope, the functional connectome of each subject was binarized, assigning a 0 to the entries with a NaN value and a 1 otherwise. The similarity value for each pair of subjects was calculated as the percentage of similarity among the binarized matrices. The process was repeated for all subjects and results were used to build an upper triangular matrix, as shown in Fig. 2.6. Each entry of the matrix was colored depending on the percentage of similarity among the corresponding subject in the row and in the column, while NaN values were assigned to mask the upper triangular elements.

2.5 Model description

In this study, regression-based models were designed and implemented at the scope of predicting how connectivity changes during motor tasks and identifying the correct subject among the others. First of all, a *degree-normalization*

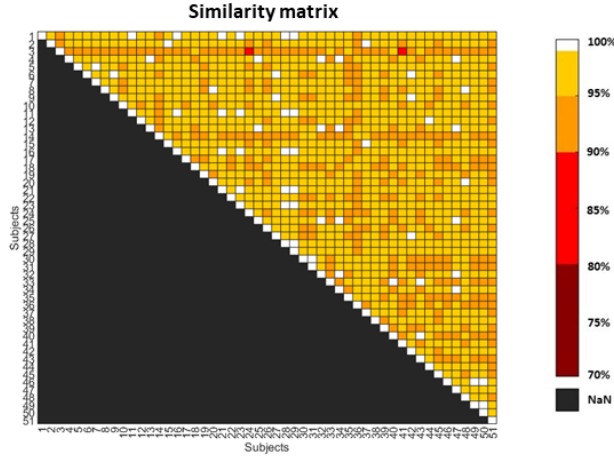


Figure 2.6: Example of similarity matrix obtained with 51 subjects.

operation (see Section 2.2.1) was applied to each matrix to reduce the excessive influence of highly connected nodes. For each subject, a mean FC matrix was computed for rest and task conditions, averaging the normalized matrices extracted from each run of rest and task, respectively. A vector was extracted from each matrix, concatenating its upper triangular elements, and provided in input to the algorithm. Inspired by Tavor’s approach [4], a regression model was designed for each subject and each link of FC. The values in the resting-state FC vectors were considered as independent variables, and the ones related to task conditions as dependent ones. The form of the model is as follows:

$$y_{nk} = x_{nk} * b_{nk} + \epsilon_{nk} \quad \begin{matrix} n = 1, 2, \dots, N \\ k = 1, 2, \dots, K \end{matrix} \quad (2.9)$$

where N and K represent the number of subjects and links; x and y the independent and dependent variables and ϵ the error term between the two variables. Regression coefficients, b_{nk} , were estimated for each link and subject separately, and then used in a LOOCV procedure to predict brain connectivity during task performance. Given any subject n , for each link in the FC matrix, the procedure consisted in averaging the regression coefficients of the remaining $N - 1$ subjects at the scope of obtaining an average coefficient

useful for the final prediction, as depicted in the following equation:

$$\overline{b_{nk}} = \frac{\sum_{i \sim n} b_{ik}}{n - 1} \quad k = 1, 2, \dots, K \quad (2.10)$$

The averaged coefficient, $\overline{b_{nk}}$, was then used for the prediction of link k of subject n , as follows:

$$\widehat{y_{nk}} = x_{nk} * \overline{b_{nk}} \quad \forall \text{ link } k \quad (2.11)$$

Finally, all the predicted values for subject n were gathered in a single array, $\widehat{Y}_n$, providing an estimation of its functional connectome during the motor task.

This algorithm was written in MATLAB using some additional tools like *FieldTrip Toolbox* [84] and *BrainNet Viewer* [85] for processing raw signals and for visualization purposes. It was first tested considering the average of the functional connectivity matrices during finger tapping (MH) and toes squeezing (MF) tasks, independently on the right or left limb, and with the whole set of brain nodes (*ALLNTW*). Afterwards, a differentiation between the right (RH) and left (LH) hand, as well as the right (RF) and left (LF) foot was performed, together with a specialization in one of the networks described in Section 1.1.1. At this scope, the original 164x164 FC matrix associated to the task and the frequency band of interest was processed in order to extract only nodes belonging to the network considered, as reported in Table 2.1.

2.5.1 Cross-network analysis

The model was also tested considering the connections between two networks. The objective was to identify a connectivity pattern among networks in order to reveal interesting properties of individual subjects and allows their identifiability. Consistently with the previous analysis, the portion of FC matrix containing links of interests was isolated and processed to reduce the influence of highly connected nodes. However, the normalization operation

described in Section 2.2.1 could no longer be applied because the extracted matrix was no more squared and symmetric since the number of networks' nodes is different. Therefore, *degree-normalization* was applied to the entire 164x164 matrix and then, only connections between nodes of interest were extracted. Once the links of interest were isolated from the normalized FC matrix, the algorithm was applied as described in the previous section.

2.5.2 Cross-task analysis

Another important analysis was carried out at the scope of verifying whether the model is specific on the particular motor task, or it can be used to model other tasks performed by the same subject. At this scope, the model was first trained with data from finger tapping and tested with data of toes squeezing. The same rationale was also applied reversing the two tasks (toes squeezing $\rightarrow$ finger tapping).

2.6 Performance evaluation

Once the predictions were estimated, it was necessary to evaluate the performance of the model to understand if the predictions were close to the actual values and if the model was able to capture the inter-individual variability among subjects.

The goodness of the model predictions was evaluated with the mean squared error (MSE). This measure was used to capture the ability of the model in predicting FC values as close as possible to the actual ones and it was computed as follows:

$$MSE = \frac{1}{K} \sum_{k=1}^K (Y_{nk} - \widehat{Y}_{nk})^2 \quad (2.12)$$

where Y_n was the vector of K real values and $\widehat{Y}_n$ contained the K predicted values.

The inter-subject identifiability, instead, was assessed by looking at the *iden-*

tiability matrix, *IDmatrix* in short [13] [4]. For each subject, the Pearson's correlation coefficient (see Section 2.2.2) was computed between its predicted connectome and the true connectome of each subject in the dataset. The results were then used to build the *IDmatrix*: a squared and non-symmetric similarity matrix of size N^2 , with N the number of subjects in the dataset. An example is shown in Fig. 2.7, where only 3 subjects were considered for simplicity. Each entry of the matrix, r_{ij} , reflects the correlation coefficient computed between the predicted connectome of subject i , in the corresponding row, and the actual FC values of subject j , in the column. For visualization purposes, a color bar from light blue to red was used to indicate the values of the correlation coefficients, from the lowest to the highest.

Although, the *IDmatrix* is a good visualization tool to investigate the per-

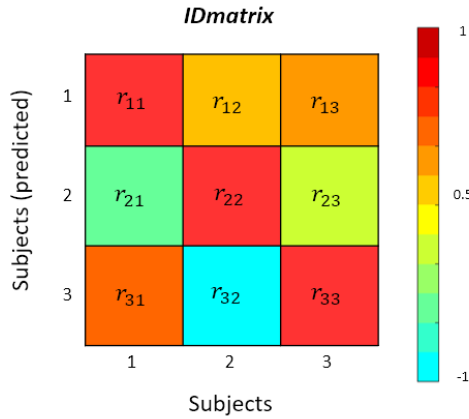


Figure 2.7: Example of a *IDmatrix* with 3 subjects.

formance of the algorithm, two metrics were extracted from this matrix to quantitatively assess the model performance.

Prediction score (*Pscore*) is the average of the diagonal elements in the *IDmatrix*. Looking at the exemplary matrix in Fig. 2.7, it is computed as follows:

$$Pscore = \frac{r_{11} + r_{22} + r_{33}}{3} \quad (2.13)$$

and it reflects the ability of the algorithm to make a good prediction of the functional connectome.

Identification rate (IDrate) [13] is an identification scoring metric, computed with the following formula:

$$IDrate = \left(\frac{\text{subjects correctly identified}}{\text{total number of subjects}} \right) * 100 \quad (2.14)$$

This metric was used to investigate the ability of the model in identifying the correct subject with respect to the others in the dataset. In particular, a subject is *correctly identified* if its value on the diagonal of the *IDmatrix* is higher with respect to the off-diagonal elements.

2.6.1 Variability associated with the dataset size

In order to understand how the different metrics behave when the size of the dataset changes, the model was applied with different sets of subjects and the performance evaluated. More in detail, the algorithm was first tested with 5 subjects randomly selected from the original dataset, then other 5 subjects were added to the previous set and the model re-trained. The procedure was repeated until the subset included all subjects. To avoid the bias in the random selection of the subjects, 20 trials were performed selecting each time a different subset of subjects. Finally, the *Pscore* and *IDrate* were evaluated for each subset of the dataset as follows:

$$Pscore_{S_n} = \frac{\sum_{t=1}^T Pscore_{S_n}(t)}{T} \quad S_n, n = 5, 10, 15, \dots, 51 \quad (2.15)$$

$$IDrate_{S_n} = \frac{\sum_{t=1}^T IDrate_{S_n}(t)}{T} \quad S_n, n = 5, 10, 15, \dots, 51 \quad (2.16)$$

where $T = 20$ corresponds to the trials performed and S_n indicates the subset of the dataset containing n random subjects.

2.7 Predicting muscular activity from brain connectivity at rest

In this section, an advanced analysis was described at the scope of predicting inter-individual differences in muscle activation during motor tasks starting from the same set of FC data as before. At this scope, the algorithm described in Section 2.5 was adapted to model the EMG data with resting-state FC information. Two EMG features were taken into consideration: RMS and ZCR (see Section 2.3.1). The EMG signal of interest was decomposed into epochs associated with movement repetitions, and the two EMG features were estimated for each segment. These values were then averaged for every single subject and a control analysis was performed at the scope of verifying the inter-subject variability across different motor tasks.

Afterwards, the model in Section 2.5 was applied exploiting FC values during motor task conditions as independent variables and one of the EMG features listed above as predictors. This preliminary study was done at the scope of solving the identifiability problem with an easier task: predicting differences in muscle activation considering FC values acquired during motor task conditions.

A step forward was made focusing on the investigation of whether individual differences in muscle activation could be predicted by FC differences while subjects are at rest. At this scope, two different attempts were performed, both considering RMS or ZCR values as dependent variables. Initially, the resting-state FC values were used as predictors, in the model described in Section 2.5, in order to identify a relationship between resting-state FC values and an EMG feature. Subsequently, a second experiment was performed considering as independent variables three graph metrics extracted from FC matrices at rest: weighted degree, average path length and efficiency (see Section 2.2.1).

Moreover, a band-limited approach was applied in both cases: the EMG signals were filtered in the frequency band of interest, before the feature extraction process. Note that this operation was performed only considering β_{low} and β_{high} frequency bands, since the pre-processing operation on EMG

signals had narrowed the frequency content in the range $[15 - 450]$ Hz (see Section 2.3).

2.7.1 Modified identifiability matrix

Also in this case, the ability of the model in identifying single subject was assessed by looking at the *IDmatrix*. However, MSE replaced the Pearson's correlation coefficients in the matrix, since the latter could no longer be calculated as the independent variables were no more vectors, but single values. Two additional modifications were applied to the *IDmatrix* in order to simplify the visualization, as reported in the example in Fig. 2.8. The color scale was inverted with respect to the previous case (Fig. 2.7), assigning a light blue color to high values and a red one to the low ones. This operation was applied to remain consistent with the colors used in the FC analysis: in this case, indeed, a subject is *correctly identified* if its diagonal element is smaller (lower error between predicted and actual EMG feature) than the off-diagonal ones. Secondly, since MSE values were very low in all conditions, a normalization operation in the interval $[0, 1]$ was applied to IDmatrices with the following formula:

$$IDmatrix_{norm} = \frac{IDmatrix - \min(IDmatrix)}{\max(IDmatrix) - \min(IDmatrix)} \quad (2.17)$$

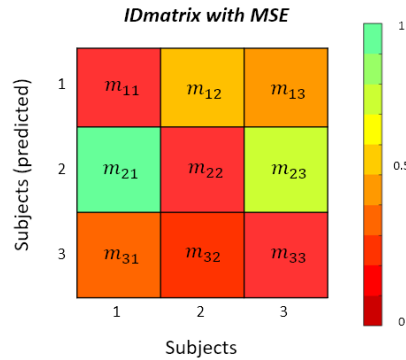


Figure 2.8: Example of a modified *IDmatrix* with 3 subjects.

Chapter 3

Results

All the analyses were performed considering different motor tasks (i.e. finger tapping and toes squeezing), frequency bands (α , β_{low} and β_{high}) and networks (Section 1.1.1). Despite the analyses were carried out for all 10 networks, this chapter reports only results obtained with the most significant networks.

3.1 Functional Connectivity variability

The FC matrices employed in this study are very different from each other, both within and between subjects. Looking at the pre-processed dataset, they differ both in the values of FC and in the structure of the matrix, or rather, in the position of NaN values (see Section 2.2). For this reason, two analyses were performed with the purpose to investigate the intra-subject (Section 3.1.1) and inter-subject (Section 3.1.2) variability of FC matrices.

3.1.1 Intra-subject similarity

From a first inspection of the connectivity matrices of a single subject, it came out that matrices associated with the same condition are different one from each other. This can be seen looking at Fig. 3.1, where FC matrices for an exemplary subject are reported during the three runs of resting-state. Each entry of the matrix was colored depending on the FC value between

the nodes in the corresponding row and column, while a white color was assigned to NaN values. To quantitatively assess the level of similarity be-

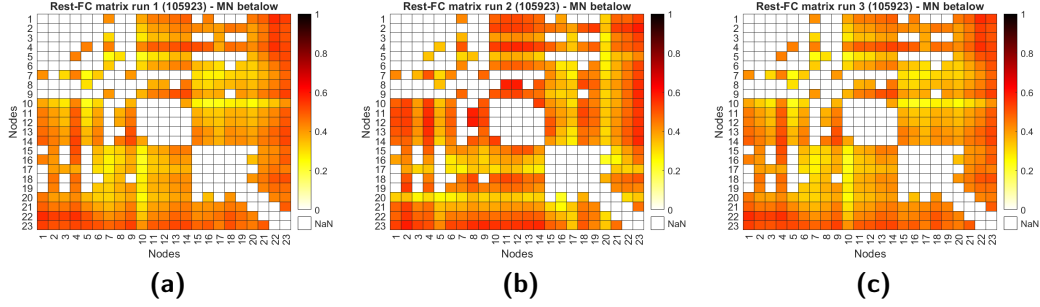


Figure 3.1: Rest FC matrices during the three runs of resting-state for subject "105923".

tween rest matrices, a consistency analysis was carried out as described in Section 2.4.1. First of all, a single value of similarity was computed for each subject with eq. 2.8. Almost all subjects show similarity values above 0.90 (with $\max = 1$). Therefore, to compare the results of different networks and different frequency bands, the values of CS_{avg} were calculated and reported in Table 3.1 for the most significant networks.

Table 3.1: Average value of cosine similarity (CS_{avg}) for the most significant networks and each frequency band.

Network	$CS_{avg} (\alpha)$	$CS_{avg} (\beta_{low})$	$CS_{avg} (\beta_{high})$
ALLNTW	0.983	0.985	0.969
CON	0.991	0.993	0.983
DAN	0.983	0.984	0.969
MN	0.983	0.986	0.977
VPN	0.984	0.987	0.974

Looking at this table, it is clear that FC values at rest are almost stable across the three rest scans, regardless of the network and the frequency band considered.

3.1.2 Inter-subject similarity

As described in Section 2.2, the entries of the FC matrices contain a numerical or a "not a number" value. While FC matrices of the same subject contain the same number of NaN values in the same position, the same does not always happen when considering different subjects. Therefore, an inter-subject analysis was conducted in order to reveal the level of similarity between subjects, focusing on the structure of FC matrices and hence on the position of NaN values. The analysis described in Section 2.4.2 was repeated for all networks and some results are reported in Fig. 3.2.

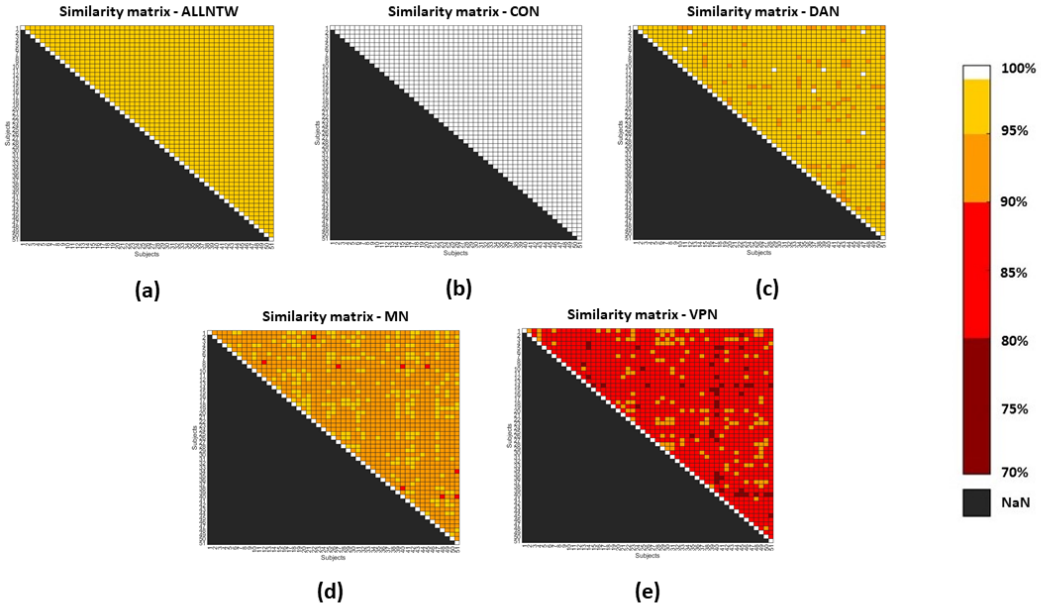


Figure 3.2: Similarity matrices for the most significant networks. (a) ALLNTW, (b) CON, (c) DAN, (d) MN, (e) VPN.

From a quick inspection of these matrices, it is evident that the matrix structure is similar among subjects, with similarity values between 90% and 99%, in almost all networks. However, two outliers are present. FC matrices in CON have the same structure for all subjects, while in VPN the similarity values are between 70% and 90% with a lower peak at 73.2%.

3.2 Modelling task from rest

After these first two analyses regarding the consistency of FC matrices, the normalization operation described in Section 2.2.1 was applied to each matrix to reduce the excessive influence of highly connected nodes and improve the algorithm performance [6]. Afterwards, the algorithm described in Section 2.5, was applied to predict task FC values from rest FC values, in every link separately. The goodness of the model was estimated by comparing the predicted values with the real ones. The estimation was performed with MSE, formula in eq. 2.12, without considering arrays' entries with NaN values in at least one of the two vectors. However, the results were not affected by this approximation because the number of NaN values was negligible with respect to the total number of links. Results are reported in Table 3.2 for MH and MF for the most significant networks. All conditions present very

Table 3.2: Average MSE (mean $\pm$ std) for the most significant networks and MH and MF in α , β_{low} and β_{high} frequency band.

Network	Band	MH	MF
ALLNTW	α	$(0.07 \pm 0.04) \cdot 10^{-5}$	$(0.07 \pm 0.05) \cdot 10^{-5}$
	β_{low}	$(0.06 \pm 0.04) \cdot 10^{-5}$	$(0.07 \pm 0.04) \cdot 10^{-5}$
	β_{high}	$(0.14 \pm 0.11) \cdot 10^{-5}$	$(0.14 \pm 0.10) \cdot 10^{-5}$
CON	α	$(3.29 \pm 2.78) \cdot 10^{-4}$	$(3.84 \pm 3.52) \cdot 10^{-4}$
	β_{low}	$(3.01 \pm 2.97) \cdot 10^{-4}$	$(3.05 \pm 2.52) \cdot 10^{-4}$
	β_{high}	$(6.19 \pm 6.58) \cdot 10^{-4}$	$(5.62 \pm 4.59) \cdot 10^{-4}$
DAN	α	$(3.83 \pm 2.71) \cdot 10^{-5}$	$(4.08 \pm 3.37) \cdot 10^{-5}$
	β_{low}	$(3.20 \pm 1.89) \cdot 10^{-5}$	$(3.22 \pm 2.91) \cdot 10^{-5}$
	β_{high}	$(6.88 \pm 6.65) \cdot 10^{-5}$	$(6.94 \pm 7.70) \cdot 10^{-5}$
MN	α	$(6.29 \pm 4.64) \cdot 10^{-5}$	$(7.03 \pm 7.83) \cdot 10^{-5}$
	β_{low}	$(4.63 \pm 3.77) \cdot 10^{-5}$	$(5.22 \pm 4.11) \cdot 10^{-5}$
	β_{high}	$(8.56 \pm 8.43) \cdot 10^{-5}$	$(9.07 \pm 8.04) \cdot 10^{-5}$
VPN	α	$(8.15 \pm 11.27) \cdot 10^{-5}$	$(8.55 \pm 10.92) \cdot 10^{-5}$
	β_{low}	$(9.71 \pm 18.75) \cdot 10^{-5}$	$(9.08 \pm 16.10) \cdot 10^{-5}$
	β_{high}	$(1.92 \pm 3.02) \cdot 10^{-4}$	$(1.91 \pm 2.58) \cdot 10^{-4}$

low values of MSE, in the order of 10^{-5} , suggesting that the prediction error

is negligible. The worst results were obtained in β_{high} and in CON, providing a hint on the identifiability performance.

3.3 Subjects identifiability

In this study, the identifiability task was accomplished thanks to the *IDmatrix* introduced in Section 2.6. A low threshold was set to -0.02 and the matrix visualized with colors from light blue to red, where the former indicates the lowest correlation coefficient ($= -0.02$), while the latter the highest one ($= 1$). Fig. 3.3 shows *IDmatrices* in MN considering MH and MF in the three frequency bands. All the *IDmatrices* are diagonal-dominant, but better

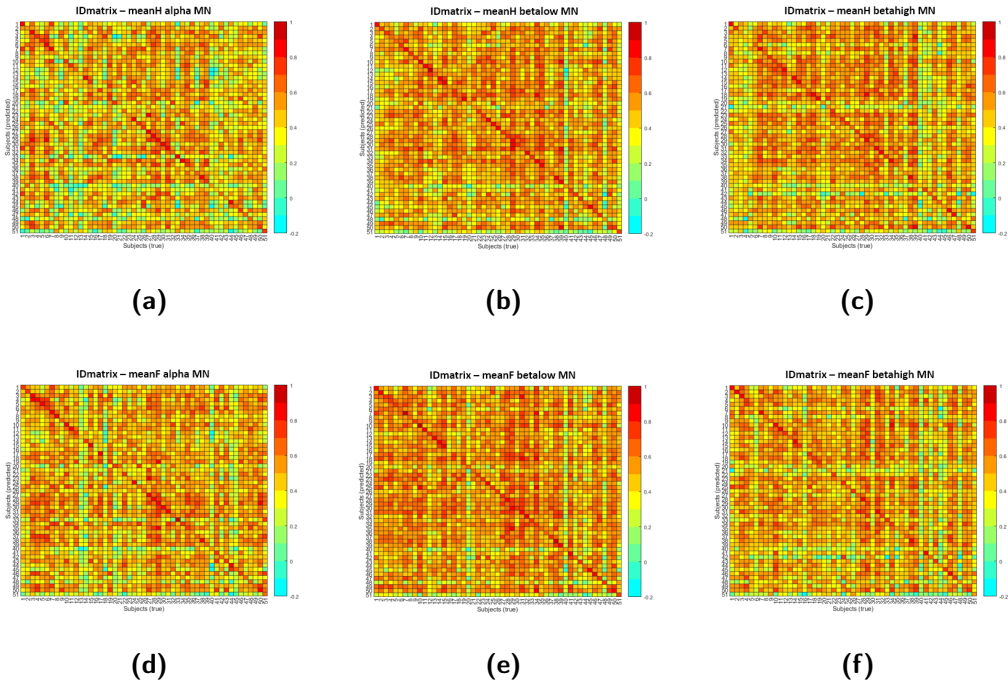


Figure 3.3: *IDmatrices* in MN for finger tapping and toes squeezing task, obtained by averaging the FC matrices computed during the execution of the motor task. (a) MH in α , (b) MH in β_{low} , (c) MH in β_{high} , (d) MF in α , (e) MF in β_{low} , (f) MF in β_{high} .

considerations can be made by looking at the *Pscore* and *IDrate* in Tables 3.3 and 3.4, where results are presented for left and right limb separately as well

as for MH and MF in MN. It is evident that the model performs well in β_{low}

Table 3.3: *Pscore* (mean $\pm$ std) calculated for LH, RH, MH, LF, RF, MF in MN. In the table are highlighted the values with a mean *Pscore* above 0.70.

Band	LH	RH	MH	LF	RF	MF
α	0.67 ± 0.15	0.65 ± 0.17	0.69 ± 0.15	0.69 ± 0.13	0.68 ± 0.16	0.72 ± 0.14
β_{low}	0.71 ± 0.10	0.71 ± 0.11	0.75 ± 0.10	0.73 ± 0.10	0.70 ± 0.12	0.76 ± 0.11
β_{high}	0.65 ± 0.13	0.66 ± 0.12	0.71 ± 0.11	0.64 ± 0.15	0.65 ± 0.13	0.69 ± 0.12

Table 3.4: *IDrate* calculated for LH, RH, MH, LF, RF, MF in MN. In the table are highlighted the values above 60%.

Band	LH	RH	MH	LF	RF	MF
α	49%	49%	54.9%	51%	49%	60.8%
β_{low}	51%	49%	62.8%	58.8%	54.9%	58.8%
β_{high}	39.2%	32.3%	49%	43.1%	39.2%	47.1%

for all the tasks, even if better results emerge considering the mean of the effectors rather than single limbs. Moreover, the average *IDrate* is around 50% in α band and 55% in β_{low} band. These values are only slightly higher than the chance level but looking at the *Pscore* it is clear that the model can predict values of FC close to the real ones, even if the identifiability metric is not as good as expected.

The same analysis was performed considering all networks described in Section 1.1.1. Fig. 3.4 shows *IDmatrices* derived with MH for the most significant networks in β_{low} . Almost all matrices are diagonal-dominant, indicating that on average the model prediction for any subject is more similar to the subject’s own connectome than to others’ connectomes. Only one exception appears in CON, where the diagonal dominance of the *IDmatrix* is not apparent. The same considerations can also be drawn looking at Table 3.5 where the *Pscore* and *IDrate* are collected for the three frequency bands. From these results, emerges that the performance of the model in VPN is very high, with a mean *Pscore* of 0.90 and an *IDrate* close to 90% in α band. Moreover, β_{high} returns the worst performance in all networks, suggesting

that this band is probably not involved in the processing of sensorimotor information.

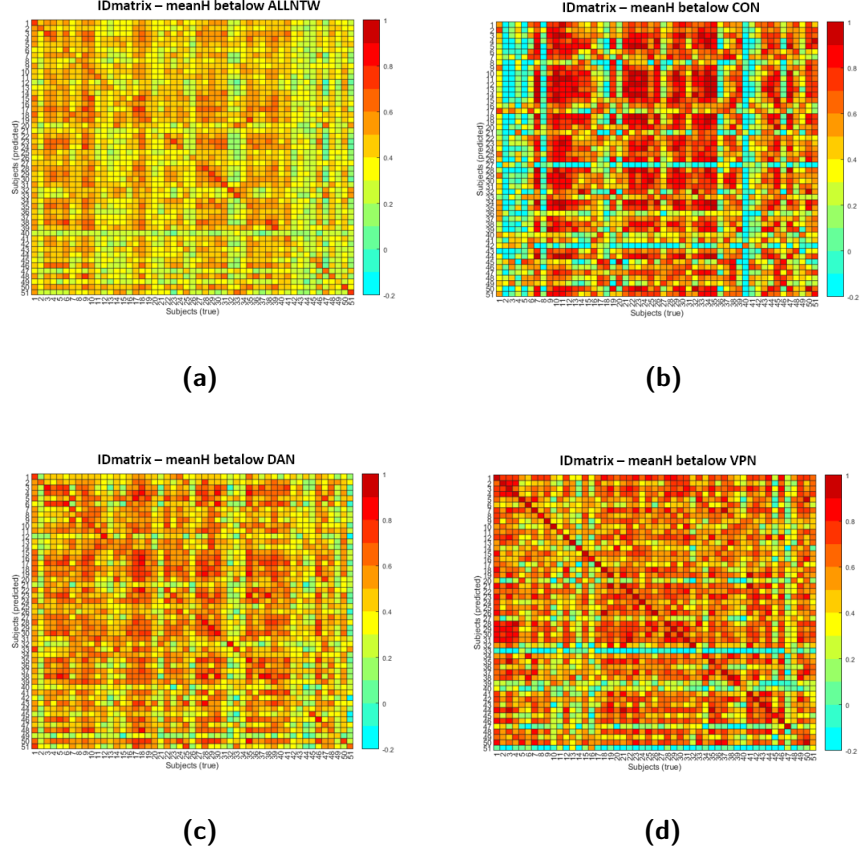


Figure 3.4: *IDmatrices* for the most significant networks in β_{low} and with MH. (a) ALLNTW, (b) CON, (c) DAN, (d) VPN.

3.3.1 Topological visualization

BrainNet Viewer toolbox [85] was used to obtain a topological visualization of brain connectomes in different conditions. This section shows two exemplary results obtained in MN during the execution of a finger tapping task in β_{low} . Two subjects were selected from the dataset and the FC values in different conditions saved as *edge* files. The two exemplary subjects were selected because of their high values of correlation between the predicted

Table 3.5: $Pscore$ (mean $\pm$ std) and $IDrate$ obtained with the most significant networks in α , β_{low} and β_{high} bands. In the table are highlighted the mean $Pscore$ values above 0.70, and the $IDrate$ values above 65%.

Network	Band	MH		MF	
		$Pscore$	$IDrate$	$Pscore$	$IDrate$
ALLNTW	α	0.58 ± 0.13	60.8%	0.62 ± 0.12	68.6%
	β_{low}	0.60 ± 0.12	47.1%	0.63 ± 0.11	58.8%
	β_{high}	0.53 ± 0.13	56.9%	0.55 ± 0.12	64.7%
CON	α	0.56 ± 0.30	13.7%	0.56 ± 0.29	15.7%
	β_{low}	0.62 ± 0.32	7.8%	0.64 ± 0.30	13.7%
	β_{high}	0.58 ± 0.32	5.9%	0.64 ± 0.30	5.9%
DAN	α	0.60 ± 0.18	45.1%	0.66 ± 0.17	49%
	β_{low}	0.68 ± 0.13	39.2%	0.66 ± 0.17	49%
	β_{high}	0.60 ± 0.17	31.4%	0.62 ± 0.15	35.3%
VPN	α	0.90 ± 0.09	88.2%	0.90 ± 0.10	92.2%
	β_{low}	0.87 ± 0.16	82.4%	0.87 ± 0.18	78.4%
	β_{high}	0.78 ± 0.25	66.7%	0.76 ± 0.26	58.8%

and actual connectomes: 0.78 and 0.89 for subject "116726" and "133019", respectively. In order to capture only the strongest connections, links with FC values above 0.01 were retained and displayed in Fig. 3.5. On the left of the figure, the connectome is visualized for both subjects during resting-state condition; in the middle, FC values are reported during finger tapping task; while, in the images on the right, is displayed the absolute difference between real and predicted connectomes during finger tapping task. From Fig. 3.5c and Fig. 3.5f emerges that the difference between predicted and actual connectome is negligible for subject "133019", but the same cannot be stated for "116726", even if it is worth noticing that all displayed links have very low values of FC. These results highlight that the model predicts a connectome very close to the actual one for subject "133019", facilitated by the fact that FC values are very similar between resting-state and task conditions, as demonstrated in Fig. 3.5d and Fig. 3.5e.

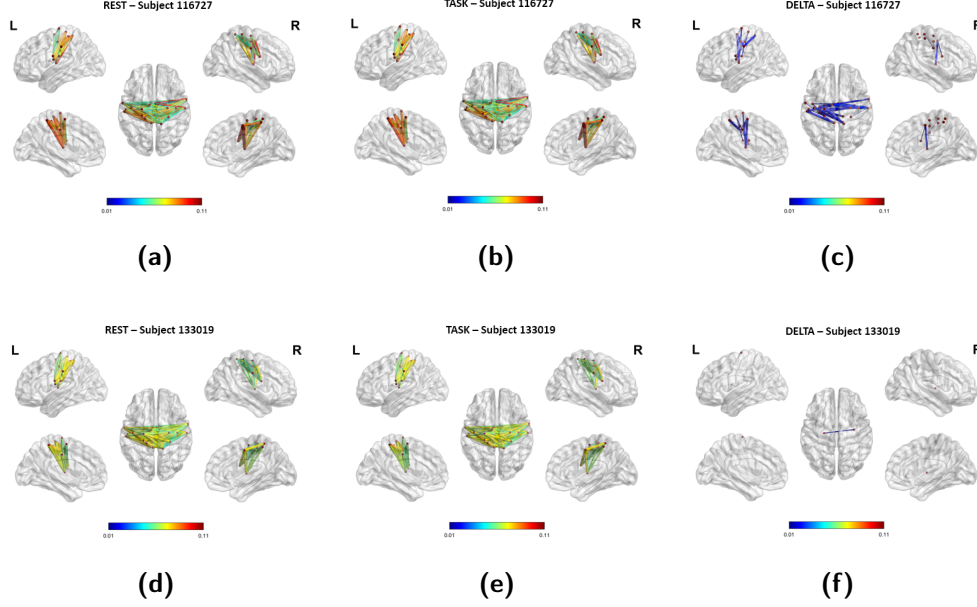


Figure 3.5: Topological visualization of FC values in MN and β_{low} band, during the execution of finger tapping considering MH. The color scale identifies the strength of FC values. (a) Resting-state subject "116726", (b) task subject "116726", (c) absolute difference between true and predicted connectome during task for subject "116726". (d) Resting-state subject "133019", (e) task subject "133019", (f) absolute difference between true and predicted connectome during task for subject "133019".

3.3.2 Between-networks identifiability

The objective of this analysis is to investigate if the implemented model is able to capture inter-individual differences in connectivity patterns between different networks. As reported in the previous literature [86] [87], MN and DAN interact a lot during cognitive, attentional and motor tasks. Therefore, a between-networks analysis was carried out considering these two networks. Quantitative results are gathered in Table 3.6 for MH and MF. Unfortunately, neither $Pscore$ nor $IDrate$ are significant in this analysis, probably due to the inability of the model in capturing inter-individual differences between the two networks. Therefore, no further tests were performed considering the interaction of the networks.

Table 3.6: *Pscore* (mean $\pm$ std) and *IDrate* for the interaction between DAN and MN in α , β_{low} and β_{high} band.

Band	MH		MF	
	<i>Pscore</i>	<i>IDrate</i>	<i>Pscore</i>	<i>IDrate</i>
α	0.56 ± 0.16	39.2%	0.61 ± 0.16	41.2%
β_{low}	0.63 ± 0.13	39.2%	0.65 ± 0.13	43.2%
β_{high}	0.54 ± 0.16	51%	0.55 ± 0.15	54.9%

3.4 Model robustness study

Some additional analyses were performed in order to determine if the model depends on the structure of FC matrices, the number of nodes in the networks, the size of the dataset and the motor task.

3.4.1 Influence of position of NaN values

As described in Section 2.2, FC matrices contain both numerical and "not a number" values. The position of NaN values in any FC matrix is specific for each subject and their number can vary from one subject to another. Therefore, it was important to verify that the ability of the model in identifying a single subject was not biased by the matrix structure, but it was dependent on the model's ability to capture individual patterns of connectivity. At this scope, two vectors (S1 and S2 in Fig. 3.6) were taken as an example to demonstrate that Pearson's correlation does not consider cells with NaN values in at least one of the two vectors. Several trials were performed adding NaN values in different positions of the two arrays. If the two vectors are perfectly equal, both in the numerical values and in the position of NaN values, their correlation is 1 (see Fig. 3.6a). The same happens if the two vectors are equals in the connectivity values but with different numbers of NaN entries (see Fig. 3.6b). If the two vectors contain different numerical values, instead, the correlation coefficient is between 0 and 1. Moreover, a shuffling operation on vectors entries does not affect the correlation coefficient, if the correspondence between elements of the two arrays is maintained, as demonstrated in Fig. 3.6c-d. Looking at these examples, it is possible to conclude that the

structure of FC matrices does not influence the performance of the model.

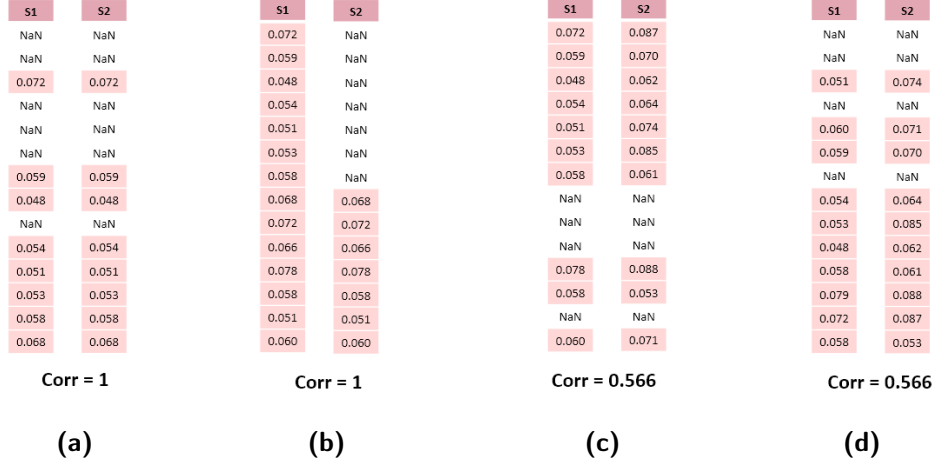


Figure 3.6: Examples of Pearson's correlation measurements with different vectors. In (a) and (b) S1 and S2 contain the same numerical values. (a) the position of NaN values is the same in the two vectors, (b) S1 has only numerical values while S2 has half entries equal to NaN. In (c) and (d) S1 and S2 have different numerical values but NaN entries are in the same position. From (c) to (d) a shuffling operation was performed maintaining the correspondence between elements of the two vectors.

3.4.2 Influence of the number of nodes on identifiability

To exclude a possible bias of the model on the number of networks' nodes, a hypothesis test was performed for each task and each frequency band. At this scope, *IDrate* values were computed for each network in Section 1.1.1 and gathered in a single vector together with the number of nodes of the network. Pearson's correlation coefficient and the associated p-value were estimated, and the latter compared with a significance threshold $\alpha = 0.05$, as described in Section 2.2.2. From a visual inspection of Fig. 3.7, it appears that the number of networks' nodes and the *IDrate* values are not linearly related, at least in β_{low} with MH. To generalize this observation to all conditions, quantitative results are presented in Table 3.7 for MH and MF in all frequency

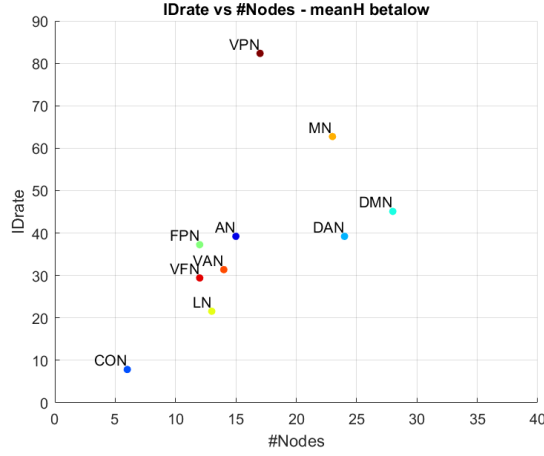


Figure 3.7: Plot number of nodes in the networks versus the IDrate values obtained in β_{low} and MH with 51 subjects. For visualization purposes ALLNTW ($IDrate = 47\%$, $\#Nodes = 164$) was not visualized.

bands. From this table, it is clear that p-values are always higher with respect

Table 3.7: Pearson’s correlation coefficients ($\mathbf{r}$) and p-values computed between $IDrate$ and number of networks’ nodes in α , β_{low} , β_{high} with MH and MF with 51 subjects.

Band	MH		MF	
	$\mathbf{r}$	p-value	$\mathbf{r}$	p-value
α	0.35	0.29	0.41	0.21
β_{low}	0.19	0.58	0.39	0.23
β_{high}	0.50	0.12	0.66	0.08

to the confidence level $\alpha = 0.05$, confirming the hypothesis that the model performance is not dependent on the number of nodes and links considered.

3.4.3 Model performance vs. dataset size

The model in Section 2.5 was initially tested with 26 subjects, before expanding the dataset to the whole pool of subjects. From a quick inspection of the performance of the algorithm, it was clear that $IDrate$ and $Pscore$ behave differently when the complexity of the dataset increases. Following

the approach described in Section 2.6.1, the model was applied on datasets of different sizes and results displayed in Fig. 3.8 for β_{low} , MN, considering MH and MF in the three frequency bands.

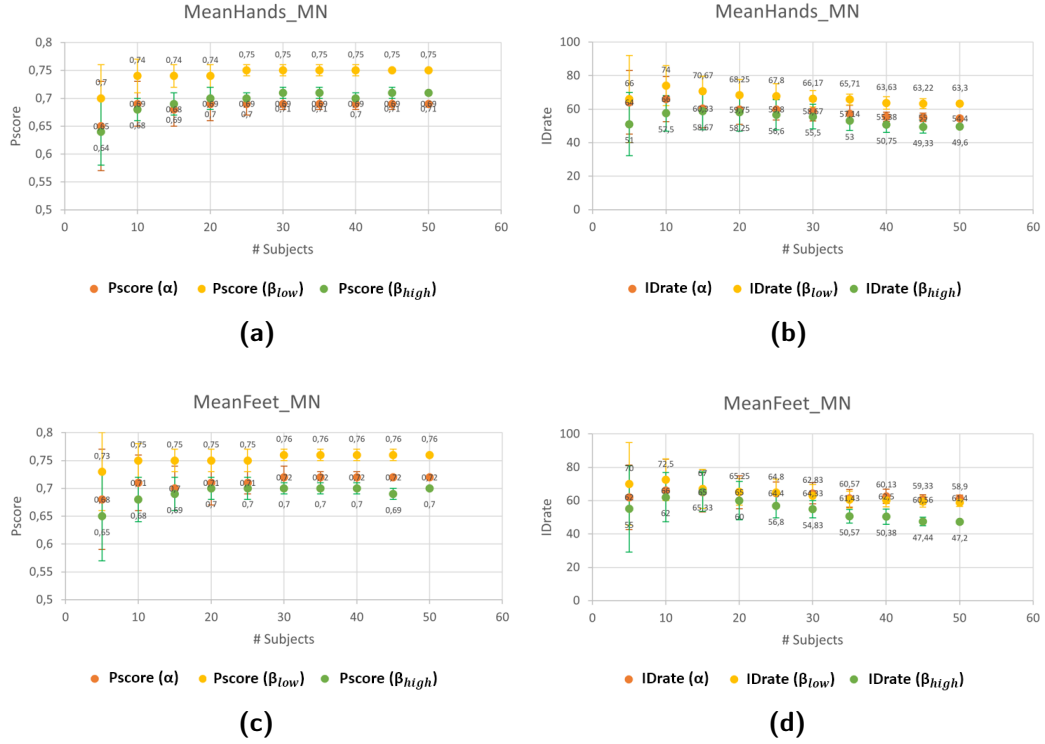


Figure 3.8: Plot the trend of $Pscore$ and $IDrate$ as the number of subjects increases. Results are reported in MN for the three frequency bands (orange for α , yellow for β_{low} and green for β_{high}). Each point represents the average value of performance and the standard deviation computed between the 20 randomizations in one frequency band. (a) $Pscore$ and MH, (b) $IDrate$ and MH, (c) $Pscore$ and MF, (d) $IDrate$ and MF.

One can expect that increasing the size of the dataset is always beneficial, but Fig. 3.8 shows that this is not always true. While $Pscore$ increases as the size of the dataset increases, the $IDrate$ starts increasing, it reaches the highest peak at around 10 subjects and then diminishes. Considering the standard deviation instead, it decreases in all cases as more subjects were added to the dataset, suggesting that approximately 40/50 subjects can be considered as a representative sample for the model.

3.4.4 Cross-task model generalization

To test the independence of the implemented model from the task considered, a cross-task procedure was applied as described in Section 2.5.2. Results of this analysis are reported in Table 3.8 in the three frequency bands of interest. Since CON led to the same results as before, i.e. training and applying the model with the same task, results are reported here only considering ALLNTW, MN and VPN.

Table 3.8: *Pscore* (mean $\pm$ std) and *IDrate* from the cross-task model analysis. *FTapping* and *TSqueezing* indicate finger tapping and toes squeezing tasks, respectively. Results obtained for three networks: ALLNTW, MN and VPN in α , β_{low} and β_{high} band. In this table are highlighted the mean *Pscore* values higher than 0.70, and the *IDrate* values above 65%.

Network	Band	FTapping $\rightarrow$ TSqueezing		TSqueezing $\rightarrow$ FTapping	
		<i>Pscore</i>	<i>IDrate</i>	<i>Pscore</i>	<i>IDrate</i>
ALLNTW	α	0.60 ± 0.12	72.6%	0.59 ± 0.13	58.8%
	β_{low}	0.62 ± 0.11	58.8%	0.55 ± 0.12	45.1%
	β_{high}	0.55 ± 0.12	66.7%	0.53 ± 0.12	54.9%
MN	α	0.70 ± 0.15	62.8%	0.69 ± 0.15	49.0%
	β_{low}	0.74 ± 0.11	58.8%	0.75 ± 0.10	60.8%
	β_{high}	0.69 ± 0.12	45.1%	0.71 ± 0.12	43.1%
VPN	α	0.90 ± 0.10	92.2%	0.90 ± 0.10	88.2%
	β_{low}	0.87 ± 0.18	78.4%	0.87 ± 0.16	82.4%
	β_{high}	0.76 ± 0.26	62.7%	0.78 ± 0.25	66.7%

From this table it is evident that, even in this case, the best performance is obtained in VPN, while worst results are obtained in β_{high} in all conditions. Moreover, it seems that the ability of the model in identifying single subject is better when the model is trained with finger tapping information and tested with FC data extracted from toes squeezing task, with respect to the opposite case.

3.5 Modelling muscle activation from rest

As described in Section 2.7, an advanced study was performed with EMG data extracted from 26 subjects of the original dataset, as EMG signals for the others were not available. The aim of the analysis consists in capturing inter-individual differences in spontaneous brain activity to explain different motor outputs. With this target, one might expect that the EMG outputs are specific enough for each subject. Therefore, the RMS and ZCR values of each subject were averaged across runs of motor task and the standard deviation was computed. Results are reported in Tables 3.9 and 3.10 for the first 5 subjects in the dataset. From a visual inspection of these tables,

Table 3.9: Inter-subject variability of RMS feature obtained averaging the RMS values between the two task runs for the first 5 subjects in the dataset in MN. Note that these values were computed considering *volt* as unit of measure of EMG signals.

Subject	LH ($\times 10^{-5}$)	RH ($\times 10^{-5}$)	LF ($\times 10^{-5}$)	RF ($\times 10^{-5}$)
105923	(7.52 ± 0.85)	(6.69 ± 0.37)	(1.31 ± 0.41)	(1.28 ± 1.18)
106521	(1.02 ± 0.45)	(1.21 ± 2.17)	(1.09 ± 3.78)	(9.32 ± 0.05)
108323	(8.34 ± 2.87)	(1.08 ± 3.15)	(1.07 ± 1.84)	(2.27 ± 0.40)
109123	(1.99 ± 0.41)	(1.65 ± 0.45)	(1.18 ± 1.32)	(1.87 ± 2.03)
113922	(1.59 ± 0.01)	(1.75 ± 1.22)	(1.22 ± 2.83)	(1.26 ± 1.41)

Table 3.10: Inter-subject variability of ZCR feature obtained averaging the ZCR values between the two task runs for the first 5 subjects in the dataset in MN. Note that these values were computed considering *volt* as unit of measure of EMG signals.

Subject	LH ($\times 10^{-1}$)	RH ($\times 10^{-1}$)	LF ($\times 10^{-1}$)	RF ($\times 10^{-1}$)
105923	(2.74 ± 0.05)	(2.87 ± 0.02)	(1.65 ± 0.13)	(2.10 ± 0.01)
106521	(2.70 ± 0.02)	(2.65 ± 0.01)	(1.36 ± 0.02)	(1.44 ± 0.02)
108323	(2.33 ± 0.07)	(2.40 ± 0.00)	(1.62 ± 0.06)	(1.64 ± 0.03)
109123	(2.73 ± 0.22)	(2.60 ± 0.21)	(3.05 ± 0.13)	(2.37 ± 0.14)
113922	(1.59 ± 0.07)	(2.35 ± 0.08)	(2.04 ± 0.05)	(1.79 ± 0.01)

it is evident that the inter-individual variability in the motor outputs is

very low, as most of the times subjects show similar EMG feature values in the same task condition. This consideration could explain why the models described in Section 2.7 did not achieve the expected outcomes. Therefore, only a few preliminary results are reported in Appendix, together with a brief explanation of them.

Chapter 4

Discussion

The present thesis aims at investigating the presence of connectivity patterns in spontaneous brain activity to predict inter-individual differences between subjects during the execution of a motor task. At this scope, different analyses were carried out to reveal the network, the frequency band and the motor task which return the best performance in the identifiability of subjects.

As highlighted in the previous chapter, FC matrices used in this thesis work showed several differences both within and between subjects. As for intra-subject consistency, it was found that FC values during resting-state conditions were similar among the three scans of resting-state, suggesting the presence of a stable organization of rest FC values that is intrinsic in the nature of each individual, and independent of external and temporal dynamics. On the other hand, the inter-subject similarity was assessed looking at the structure of FC matrices, and hence the position of NaN values in different subjects. Almost all subjects in all networks reported high values of similarity, even if two outliers were present. In CON, all subjects were characterized by the same structure of FC matrices, probably facilitated by the low number of nodes included in this network and their location far from each other, as depicted in Fig. 1.3 on page 11 and Table 2.1. The opposite happened in VPN. However, the differences in the structure of FC matrices do not affect the prediction because the model does not take into consideration the topological information in the functional connectome. This is

in contrast with some previous studies which exploited fMRI to relate rest FC values with brain activation during motor tasks [3] [4]. Tavor et al. [4], for example, developed a model to predict qualitative differences between subjects in terms of shape, position, size and topography of activations obtaining an average *Pscore* value of 0.52 and 0.56 with finger tapping and toes squeezing tasks, respectively. However, the use of MEG data to predict FC information makes the topology prediction a very challenging task that has not been investigated in this work.

The core of the analysis consisted in the study of inter-individual differences in functional connectomes at the scope of predicting task-evoked FC values and identify one subject from the others during the prediction. Looking at the metrics used for the evaluation of the performance, it was clear that the worst results were obtained in CON: the model was not able to identify individuals from the pool of subjects (*IDrate* < 15.7%) nor to make predictions close to the actual values (*Pscore* < 0.64). This is not so surprising, as CON is a high-level network, usually associated with cognitive tasks, rather than sensorimotor ones [88]. The best results, instead, were obtained with VPN. This is quite unexpected since the performed task is motor, not visual, but it may suggest that the model effectively capture an individual fingerprint independent of the performed task. This statement is even more trustworthy thanks to the fact that previous studies have demonstrated that the visual network remains almost stable between rest and task conditions [89] [90], suggesting that fingerprints are inherent to the individuals and independent of volatile factors, like task strategy and compliance [4]. Moreover, considering the results reported for MN, it came out that the performance of the model with single limbs is worse than the one obtained averaging the FC matrices related to the movement of the left and right effectors. This could be due to the fact that the average FC matrix has the property of highlighting the strongest connectivity patterns with respect to the specificities, i.e. the differences in the single limb. Therefore, the model can better capture the strongest patterns and thus identify the subjects by looking at the FC matrix obtained considering both effectors together. However, also considering MH and MF in MN, the average *IDrate* value was only slightly higher than

the chance level, even if *Pscore* clearly shows that the model prediction was similar to the actual one. This can be explained by looking at the results reported in Section 3.4.3. While *Pscore* represents the ability of the model in predicting FC values as similar as possible to the actual ones (regression), the IDrate reflects the model’s capacity in classify different subjects (classification). As expected, the increased complexity of the dataset makes the identifiability task more difficult, even if the prediction becomes closer to the actual values.

Finally, considering the comparison between the frequency bands, β_{high} returned the worst performance in all networks, while results with α and β_{low} were much better ($IDrate_{\alpha} > 49\%$, $IDrate_{\beta_{low}} > 49\%$ in MN). This outcome reveals that β_{high} is not involved in the identification of connectivity fingerprints during the execution of a specific motor task. This is consistent with previous studies [91] [92] that have demonstrated that while β_{high} band is associated with cognitive functions, β_{low} is more involved in the processing of sensorimotor information.

Conclusions

When asked to perform the same task, different individuals exhibit markedly different patterns of brain connectivity. It is thought that this variability was mainly associated with differences in gross brain morphology and volatile factors such as task strategy and compliance [4]. Recently, researchers have investigated the possibility that individual differences in brain connectivity are inherent features of individuals that can be captured by looking at task-independent measurements collected at rest [4] [93]. To this end, it is common to model the brain as a network composed of nodes (brain connectome), interconnected by functional links [6], which represents the statistical interdependencies between brain regions [13]. It is now well-established [94] that the continuous repetition of a task produces connectivity patterns that leave a trace in the spontaneous brain activity. These patterns are specific to each individual and they can be considered as "fingerprints" for subject identification across rest and task conditions [3]. Therefore, few studies have started to explore connectome fingerprinting in different functional neuroimaging modalities, such as fMRI, and to explore the extent to which these fingerprints can be used to identify subjects across runs of rest and task conditions [2] [3] [4].

This work addressed these open questions with MEG data. It consists in the development of a machine learning algorithm to capture inter-individual differences at rest and to predict differences in motor tasks. In this work, two simple motor tasks were considered for the analysis: finger tapping and toes squeezing. The purpose of the model is to predict how the functional connectome changes during task conditions and how the inter-individual differences in the spontaneous brain connectivity can be exploited to identify

single subjects. Results demonstrated that, on average, the model prediction for any given subject is more similar to the subject’s own connectome than to other subjects’ connectomes. This suggests that resting-state connectivity alone is sufficient to predict individual differences in task-evoked FC in different networks and frequency bands with the best results obtained in β_{low} and finger tapping task, considering the average of the FC matrices, independently on the right or left limb. Moreover, VPN is the network which returns the best performance supporting the hypothesis that the individual fingerprints captured by the model are not dependent of the performed task. Afterwards, a few analyses were carried out at the scope of investigating the possibility of finding a relationship between brain FC at rest and EMG activation during motor task execution. For this purpose, different experimental models were developed relating features extracted from EMG data (i.e. RMS and ZCR) with resting-state FC information. However, none of them was able to predict inter-individual differences in muscular outputs by only looking at the differences in the spontaneous brain activity.

In the future, new tests should be done considering different EMG features. In particular, it could be interesting to develop a multi-class classifier in order to find the EMG features that better capture the inter-individual differences in muscle activation and use them in the model for the identifiability task. A further improvement could be made by changing the architecture of the model at the scope of increasing the algorithm performance. By increasing the complexity of the dataset, a multivariate or a deep learning approach should be developed in order to use multiple variables for forecasting: FC information, EMG features and behavioural data could be used as predictors to capture stronger inter-individual differences and hence improve the identifiability task. Moreover, the model should be tested with more complex tasks, not only related to the sensorimotor domain.

Although the limitations and the future improvements highlighted above, this thesis work found a way to predict inter-individual differences in brain connectivity, identifying a regression model between resting-state and task conditions. This framework has great implications in medicine and neuro-

science because it opens novel opportunities in the development of personalized medicine and neuroprostheses. The hope is that future research will benefit from this preliminary study to predict brain connectivity with MEG data.

Appendix

Inter-subject identifiability with EMG signals

All the analyses with EMG data were carried out considering 26 subjects of the original dataset because the EMG signals from the other subjects were not available. Moreover, results are reported only in MN, for simplicity. The first implemented model was described in Section 2.7 and it aims to predict EMG features (RMS or ZCR) looking at FC values at rest. Fig. 4.1 shows the *IDmatrices* for the four limbs in β_{low} and MN, with both EMG features.

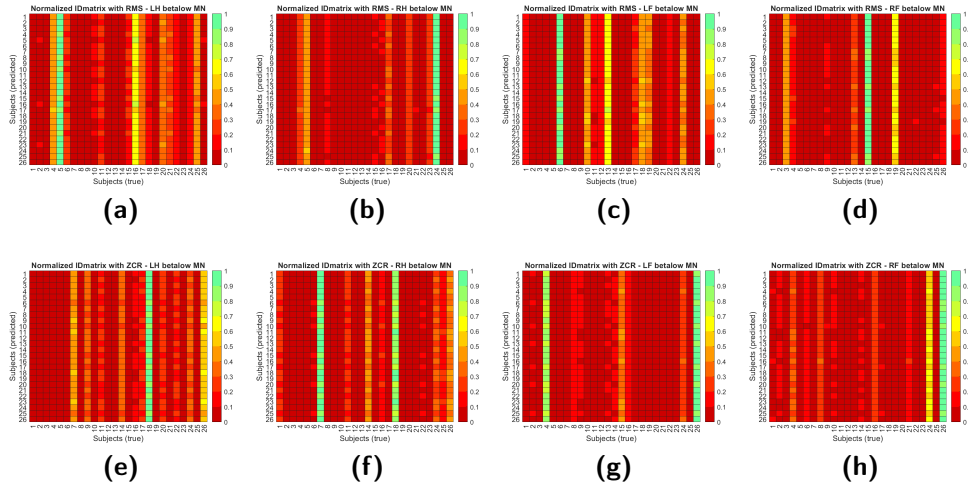


Figure 4.1: Plot the normalized *IDmatrices* obtained with 26 subjects in β_{low} and MN relating EMG features with FC values at rest. (a) LH with RMS feature, (b) RH with RMS feature, (c) LF with RMS feature, (d) RF with RMS feature. (e) LH with ZCR feature, (f) RH with ZCR feature, (g) LF with ZCR feature, (h) RF with ZCR feature.

From these plots, it is evident that none of the *IDmatrices* is diagonal-dominant, suggesting that the model is not able to identify single subjects in the dataset. What happens, instead, is that approximately a couple of subjects for each condition shows high values of MSE on the columns of the *IDmatrices* because their actual feature value is far from all the predicted ones (e.g. see subjects 5 and 16 in Fig. 4.1a). This phenomenon is rather strange and it has been investigated with a few additional analyses. In particular, the values of FC at rest and the EMG features of these "outliers" were compared with the ones of the other subjects, in order to verify if some substantial differences are present. However, nothing emerged from this study and so it was not possible to explain this unexpected phenomenon.

In order to identify a differentiation among the three frequency bands, a band-limited approach was applied as described in Section 2.7. Results are reported in Fig. 4.2 considering toes squeezing task with left foot in β_{low} and β_{high} . These matrices suggest that the model is not able to recognize individual subjects.

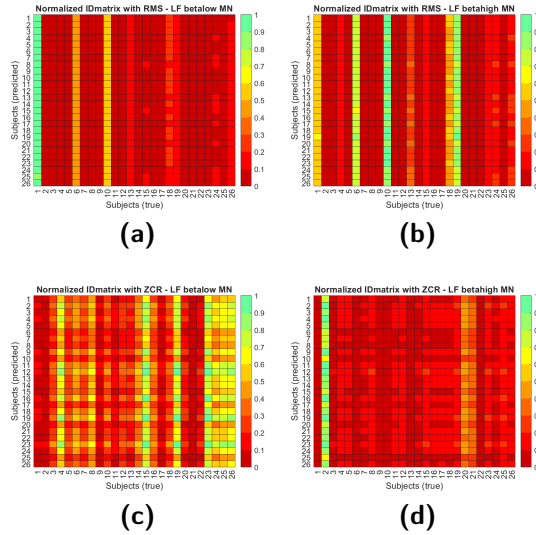


Figure 4.2: Plot the normalized *IDmatrices* obtained with 26 subjects during toes squeezing with left foot in β_{low} and β_{high} band and MN. (a) LF, β_{low} , RMS, (b) LF, β_{high} , RMS. (c) LF, β_{low} , ZCR, (d) LF, β_{high} , ZCR.

Finally, a second attempt was performed relating graph metrics extracted

from FC matrices at rest with EMG feature values (see Section 2.7). Results are reported in Fig. 4.3 for the four limbs in β_{low} and MN, both with RMS and ZCR.

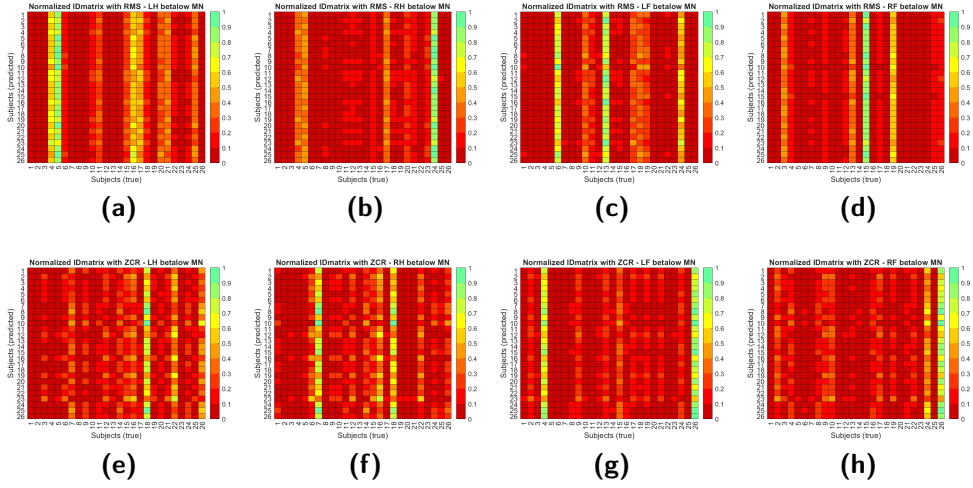


Figure 4.3: Plot the normalized *IDmatrices* obtained with 26 subjects in β_{low} and MN relating EMG features with graph metrics extracted from FC matrices at rest. (a) LH with RMS feature, (b) RH with RMS feature, (c) LF with RMS feature, (d) RF with RMS feature. (e) LH with ZCR feature, (f) RH with ZCR feature, (g) LF with ZCR feature, (h) RF with ZCR feature.

Unfortunately, also in this case, the model is not able to discriminate single subjects: *IDmatrices* appear more scrambled than before, but the diagonal dominance is still not present.

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